Unit-3
Special Purpose Diodes

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3.1 ZENER DIODE :

We saw that an electric field is produced when PN junction is reverse biased. The intensity of the field increases with the increase in the reverse voltage. The covalent bonds are broken. At some reverse voltage there is sharp increase in the reverse current and it is called the zener break down. It is seen from the characteristic that even though the current through the diode changes, the voltage across the diode remains constant. This is called the zener effect. If this reverse current is limited, the diode does not heat up. The voltage at which the zener break down occurs can accurately be decided by adding the impurity in certain definite proportion while manufacturing. This type of diode is called the zener diode. Its symbol is shown in Fig. 3.1.

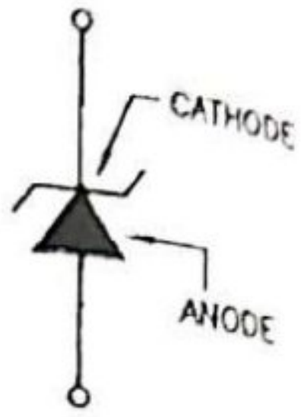


FIG. 3.1 : SYMBOL OF ZENER DIODE

Zener diode is always operated in reverse bias condition i.e. its anode is connected to the negative terminal and cathode is connected to the positive terminal of the battery. Zener diodes are manufactured in the voltage ratings from 2.4 V to 200 V and more and the power ratings of 150 mW to 50 W.

3.1.1 Characteristic :

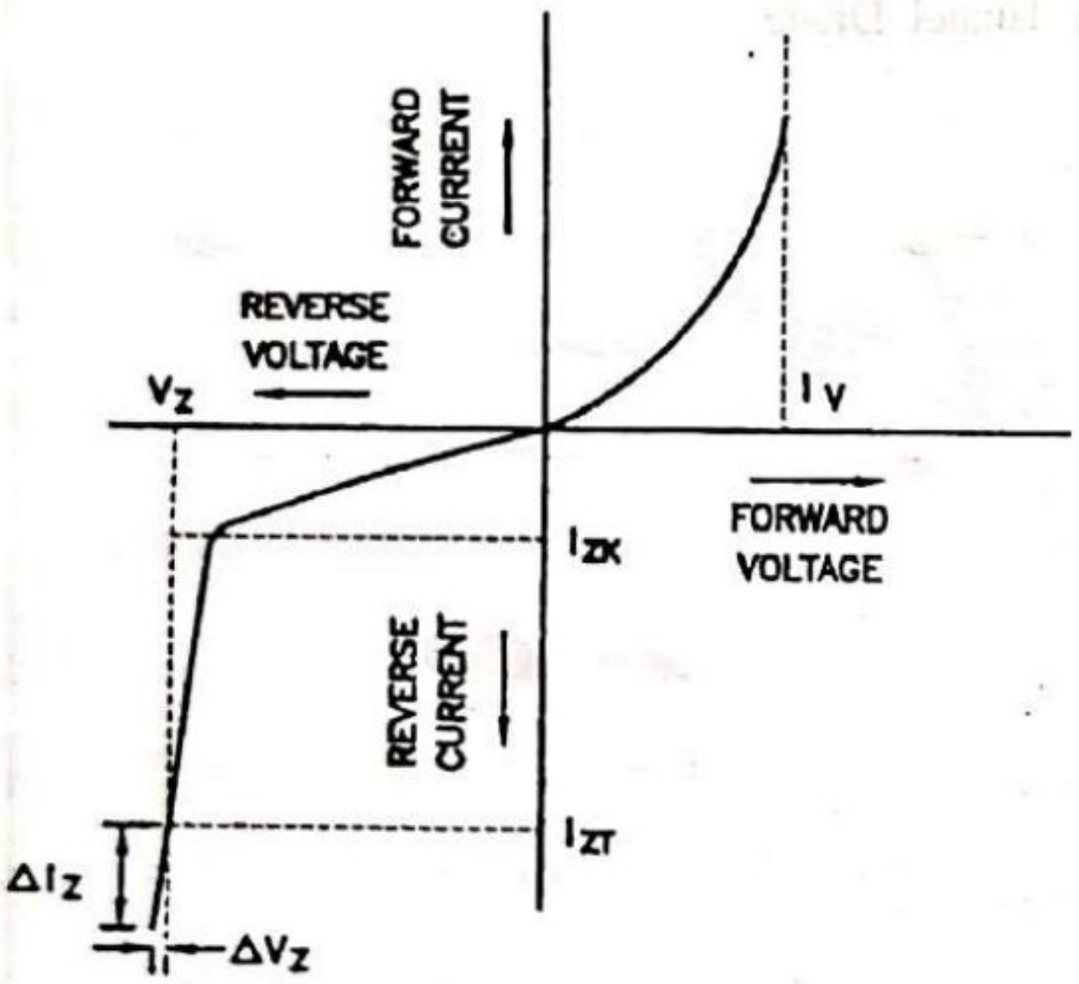


FIG. 3.2 : CHARACTERISTIC OF ZENER DIODE

Characteristic of zener diode is shown in Fig. 3.2. It is seen from the reverse characteristic that the voltage drop across the zener diode is not dependent on the current flowing through it. It means that even though the current through diode changes, the voltage drop across it remains almost constant.

V_Z is known as the zener voltage and it is specified for certain test current I_{ZT}. I_{ZK} is known as the zener knee current. Minimum of knee current I_{ZK} should be passed through the zener diode to bring it into action.

3.1.2 Operation :

One important application of the zener diode is as the voltage regulator. When load is connected to the rectifier, the d.c. load voltage changes with the change in the input a.c. voltage applied to the rectifier or with the change in the load current or both. This d.c. voltage across the load may be kept almost constant by making use of the zener diode.

Zener diode with series resistor is connected in between the d.c. output terminals of the rectifier and the load. The circuit is shown in Fig. 3.3.

Let, V_i = input voltage

V_o = output voltage

I_L = load current

I_Z = zener current

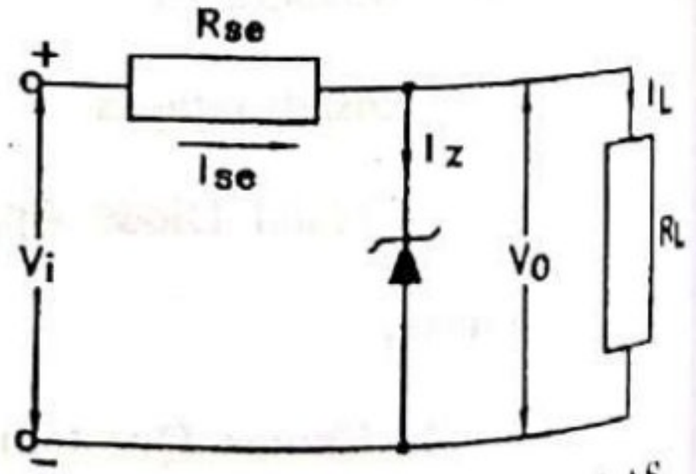


FIG. 3.3 : ZENER DIODE AS VOLTAGE REGULATOR

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I_{se} = total current

R_{se} = series resistor

R_L = load resistor

$I_{se} = I_L + I_Z$

Voltage drop Ise Rse occurs in the series resistor.

$$V_i = V_o + I_{se} R_{se}$$

$$V_i = V_o + (I_L + I_Z)R_{se}$$

Vo is zener voltage. Zener diode adjusts its current such that its voltage remains constant irrespective of the changes in the input voltage or in the load current.

If the load current remains constant and if the input d.c. voltage increases, the output voltage tries to increase but the zener diode takes more current. So current I_{se} increases. So voltage drop I_{se} R_{se} increases and the output voltage does not increase and remains constant. Similarly if the load current remains constant and if the input voltage decreases, the output voltage tries to decrease, but this time zener diode takes less current. So total current decreases and so series voltage drop decreases and output voltage V_o remains to the same value.

Now let us try to understand its action when the input voltage remains constant and the load current changes.

If the load current increases with the constant input voltage, the total current tries to increase. So the output voltage tries to decrease. But it does not happen so. Because the zener current decreases so the total series current remains constant, so the voltage drop across the series resistor remains constant, with the result that the output voltage remains constant.

Similarly with the input voltage remaining constant and the load current decreases, the zener current increases to make series current constant. So the output voltage remains constant.

3.1.3 Applications :

Various applications of zener diode are as shown below :

1. As voltage regulator
2. As reference voltage device
3. In constant current source with the transistor
4. In over voltage protection and
5. In clipping and clamping circuits

3.2 PHOTO DIODE :

When conventional PN junction diode is reverse biased, reverse saturation current flows due to the minority charge carriers (electrons in P-type and holes in N-type). More covalent bonds are broken and minority charge carries increase with the increase in temperature, so the reverse current increases. When light is thrown on the exposed PN junction then also the reverse current flows due to the increase in the minority charge carriers as a result of the light energy. Photo diode works on this principle.

3.2.1 Construction :

In photo diode silicon or germanium PN junction is packed in a plastic capsule. On one end a lens system is formed such that light is focused on the junction. Other portion is made opaque. Leads are taken out from P and N materials. Construction is shown in Fig. 3.4 (a) and symbols are shown in figure (b) and (c).

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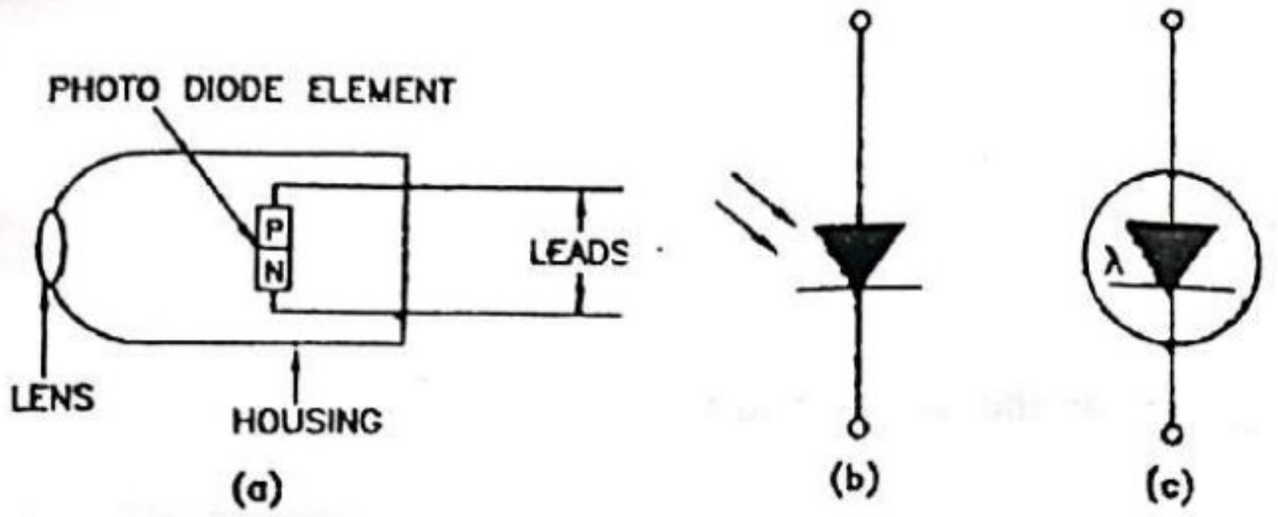


FIG. 3.4 : PHOTO DIODE (a) CONSTRUCTION (b) SYMBOL (c) ALTERNATE SYMBOL

3.2.2 Working :

Photo diode is given reverse bias. When no light falls on the junction, very small reverse current flows. This small reverse current flows as a result of breaking of covalent bonds due to the temperature. When light falls on the junction, minority charge carriers increase and reverse current increases. Value of the reverse current increases with the light. Magnitude of the reverse voltage does not affect the value of the reverse current. Reverse current remains the same even if the reverse voltage is made zero because barrier potential works as reverse bias and the reverse current continues to flow.

When the diode is given ,some forward voltage, forward current flows, which is in opposite direction to the reverse current so that reverse current decreases. And when the value of the forward current becomes equal to the reverse current, the net current becomes zero.

3.2.3 Characteristics :

Characteristics of photo diode are shown in Fig. 3.5. It is seen that most part of the characteristic is in the third quadrant, where-as some part is in the fourth quadrant.

Graphs of reverse voltage v/s reverse current are drawn for different constant values of illumination. There is small reverse current even when no light is thrown. This curve is known as the dark characteristic. This current flows as a result of charge carriers produced due to breaking of covalent bonds at room temperature. When the light is thrown on the junction, value of reverse current increases. Curves A, B, C, D are obtained for different levels of illumination.

When the reverse voltage is made zero, there is no appreciable decrease in the reverse current. When some forward voltage is applied, forward current flows, which is in opposite direction to the flow of reverse current. So net reverse current decreases. And when the value of the forward current becomes equal to the reverse current, net current becomes zero.

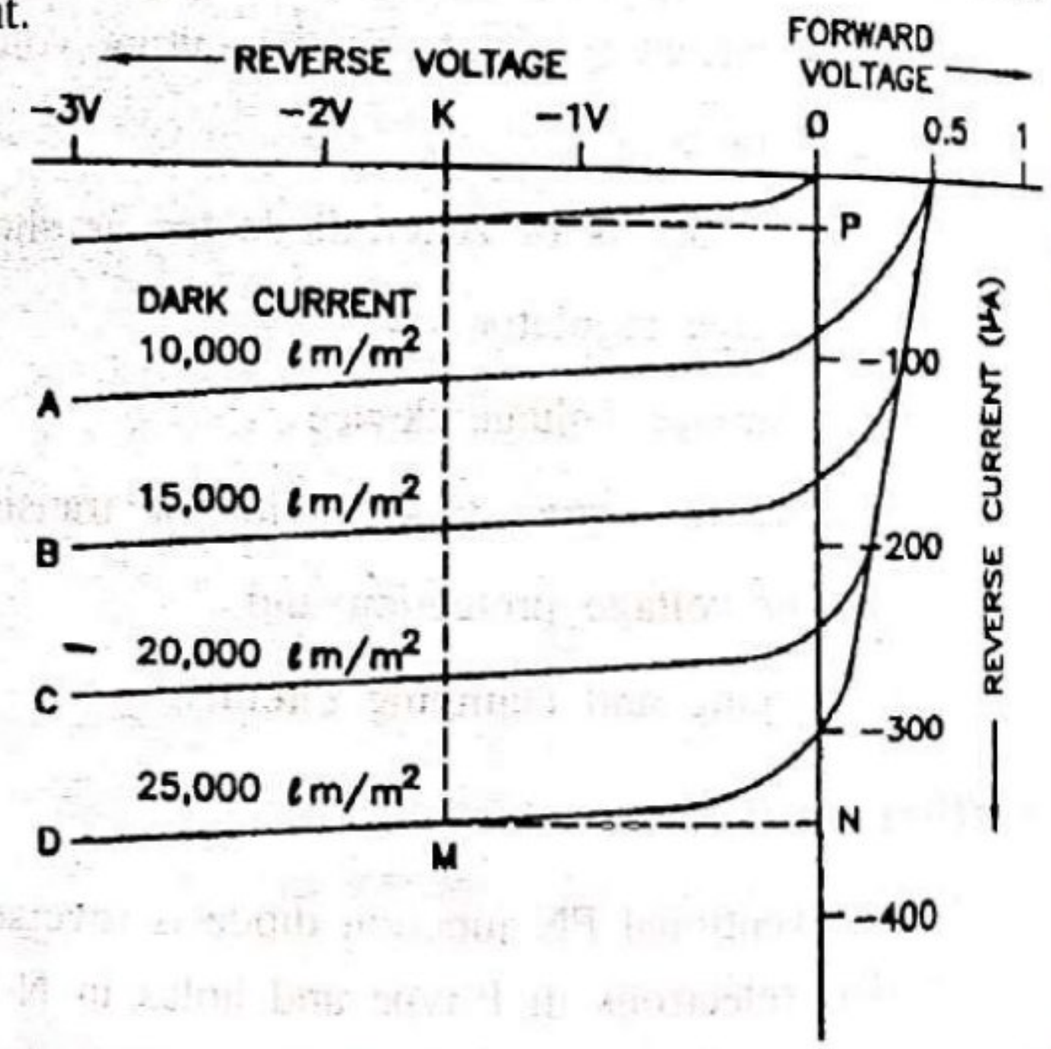


FIG. 3.5 : CHARACTERISTICS OF PHOTO DIODE

Value of dark current is 25 μA for the reverse voltage of –1.5 V, where-as for the value of this reverse voltage, the value of the reverse current is 350 μA for the illumination of 25000 lm/m^2.

3.2.4 Advantages :

1. Response time is fast.
2. Size is small.

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Disadvantages :

1. Its active area is small.
2. When the temperature is increased, increase in dark current is more.
3. It requires bias voltage for its operation.
4. It requires amplifier as the current is small in magnitude.

Applications :

Various applications of photo diode are listed below :

1. It is used in high speed switching and counting up to 1 MHz frequency.
2. It finds use in optical communication.
3. It is used in radar tracking system and numerical controlled machines as shaft encoder.
4. It is used in punched card reader and punched tape reader in computer.
5. It is found useful in brushless d.c. motor.

LIGHT EMITTING DIODE :

When PN junction is forward biased, electrons in N-region combine with the holes in P-region. Free electrons in the conduction band and holes are in the valence band. Thus electrons go to the valence band from the duction band that is electrons go to the lower energy level from the higher energy level. So when recombination electrons .and holes occurs, energy is radiated in the form of heat and light. In light emitting diode arrangement made such that the energy radiated is in the form of light. Colour of the light radiated depends upon the material the LED.

Construction :

N-type epitaxial layer is grown on the subtrate. P-region is ,formed over it by the diffusion process. combination of the charge carries take place in P-region. So p region is kept on the upper side. Lead is taken t from the P-type material such that the light is not obstructed. A layer of gold film is kept below N-layer that the light emitted is reflected. In Fig. 3.6(a) is shown the construction and in (b) and (c) the general appearance d in (d) is shown the symbol.

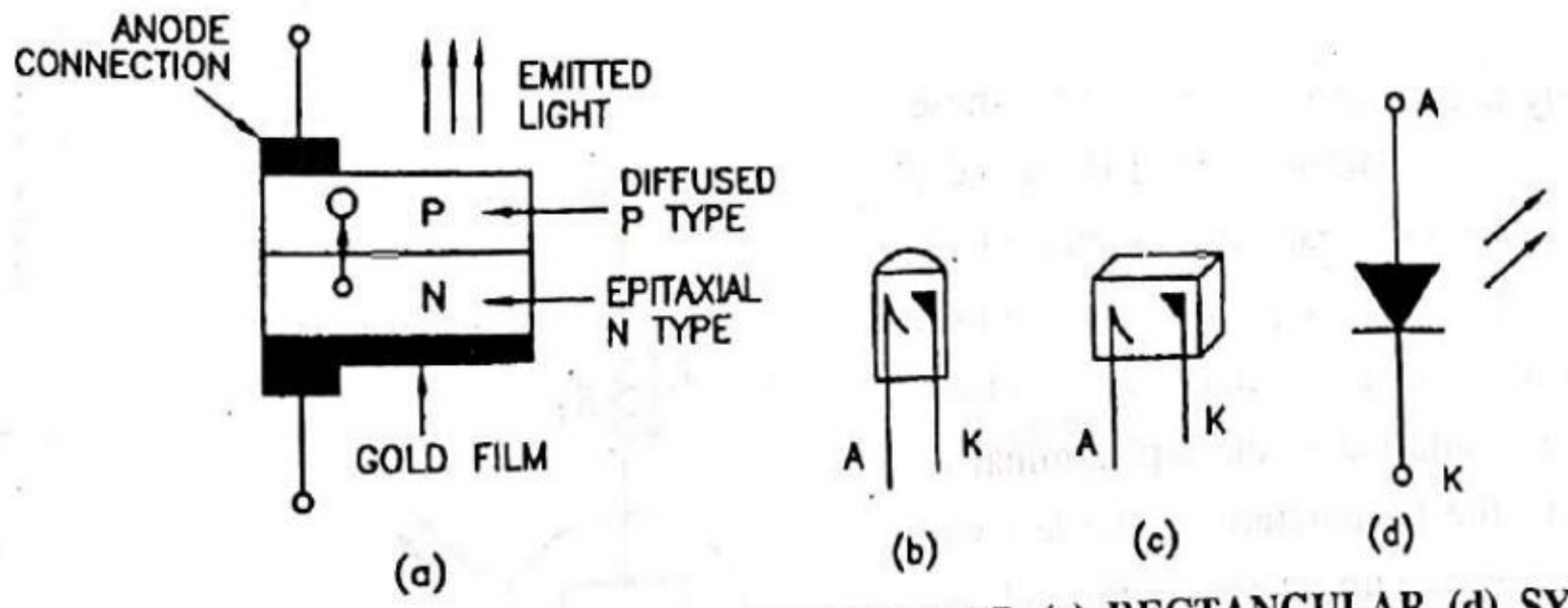


FIG. 3.6 : LED (a) CONSTRUCTION (b) ROUND LED (c) RECTANGULAR (d) SYMBOL

In the symbol, the arrows point to the outward side which indicate that the light is emitted (in photodiode e arrow heads are toward inside).

Semiconductor material used is semi-transparent. It is gallium arsenide (GaAs) or gallium arsenide phosphide GaAsP) or gallium phosphide(GaP). Red or amber (yellow) light is radiated when GaAsP is used. Green light

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is radiated with GaP and invisible infra red radiation comes out when GaAs is used. Round LEDs are available with 3 mm or 5 mm or more in diameter. Rectangular type LEDs are also available which are used in bar graph indicators.

3.3.2 Working :

LED is forward biased to bring it into action. Minimum of 10 mA current should be passed through it. Value of current can be kept between 10 mA to 25 mA. Due to the forward bias, electrons from N-type material go from the conduction band and recombine with the holes in the valence band of P-type material. So energy in the form of light is radiated.

3.3.3 Advantages :

1. Long life.
2. Can be operated at low voltage.
3. Switching is fast.

3.3.4 Disadvantages :

1. Power consumption is more compared to LCD type display.
2. Display becomes dim in the ambient light

3.3.5 Applications :

The following are some applications of LED.

1. As pilot indicator.
2. In opto-isolators.
3. In optical communication.
4. In large numerical displays (in this clusters of LEDs are used).
5. Rectangular LEDs are used in bar graphs.

3.4 MULTICOLOUR LEDs :

A LED that emits one colour when forward biased and another colour when reverse biased is called a multicolour LED.

One commonly used schematic symbol for these LEDs is shown in Fig. 3.7. Multicolour LEDs actually contain two pn junctions that are connected in reverse-parallel i.e. they are in parallel with anode of one being connected to the cathode of the other. If positive potential is applied to the top terminal as shown in Fig. 3.7(i), the pn junction on the left will light. Note that the device current passes through the left pn junction. If the polarity of the voltage source is reversed as shown in Fig. 3.7(ii), the pn junction on the right will light. Note that the direction of device current has reversed and is now passing through the right pn junction.

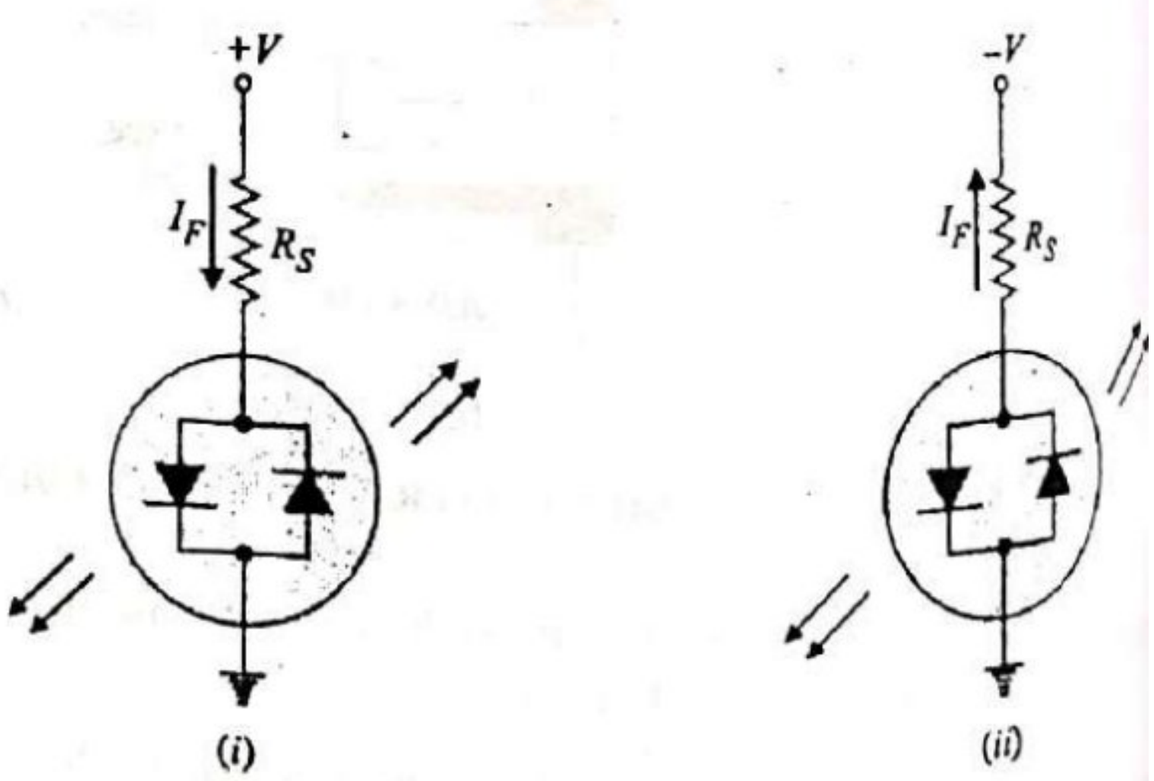


FIG. 3.7

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Multicolour LEDs are typically red when biased in one direction and green when biased in the other. If a multicolour LED is switched fast enough between two polarities, the LED will produce a third colour. A red/green LED will produce a yellow light when rapidly switched back and forth between biasing polarities.

3.5 SCHOTTKY DIODE :

One serious drawback of junction diode is that its reverse recovery time is more. That it takes time to turn off due to the stored charge. Hence ordinary junction diode can not be used for applications at high frequencies above 10 MHz. In point contact diode we saw that due to very small area of PN junction and due to the absence of depletion region, the junction capacitance becomes less of the order of 0.1 to 1 pF. So it is suitable for applications at high frequencies as high as 10 MHz and more.

3.5.1 Construction :

Schottky diode is the extension of the point contact diode. In this the metal semiconductor interface is a surface instead of point contact. It consists of a metal like silver, gold or platinum on one side of the N-type semiconductor as shown in Fig. 3.8.

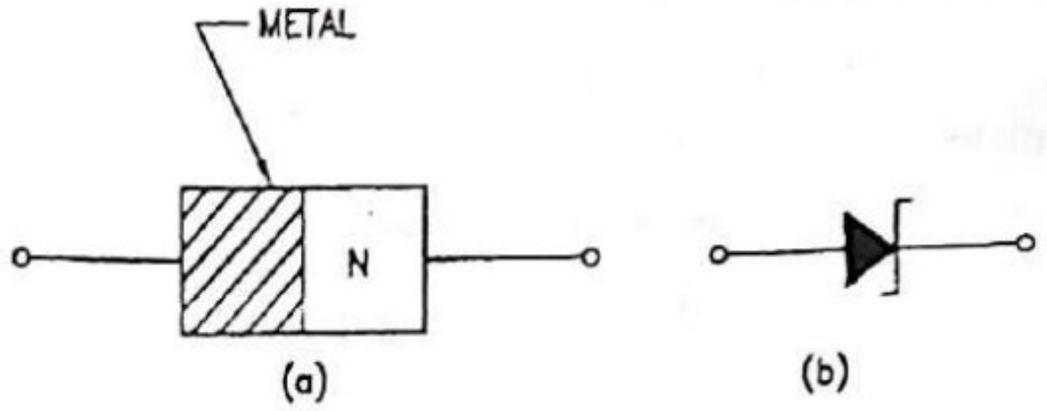


FIG. 3.8 : SCHOTTKY DIODE
(a) CONSTRUCTION (b) SYMBOL

When a Schottky diode is unbiased the electrons on metal side are in larger orbits than free electrons on N-side. This difference in the size of the orbits is called the Schottky barrier. Since greater number of charge carriers are available sufficient current is available at lower voltages and so the cut in voltage is low and the reverse saturation current is more. There are no holes in metal and also there is absence of the depletion region, there is no charge storage. So, reverse recovery time is very small or zero. Also the junction capacitance is very low and the Schottky diode can switch off faster than the bipolar diode. Schottky diode is a unipolar device because it has electrons as majority carriers on both the sides of the junction.

Schottky diode is also called the hot carrier diode because when it is forward biased, conduction electrons on N-side attain sufficient energy to cross the junction and enter the metal. As these electrons enter the metal with large energy .these are called the hot carriers.

Since the contact area between the metal and semiconductor is large, it results in lower forward resistance and so the noise is reduced.

3.5.2 Characteristics :

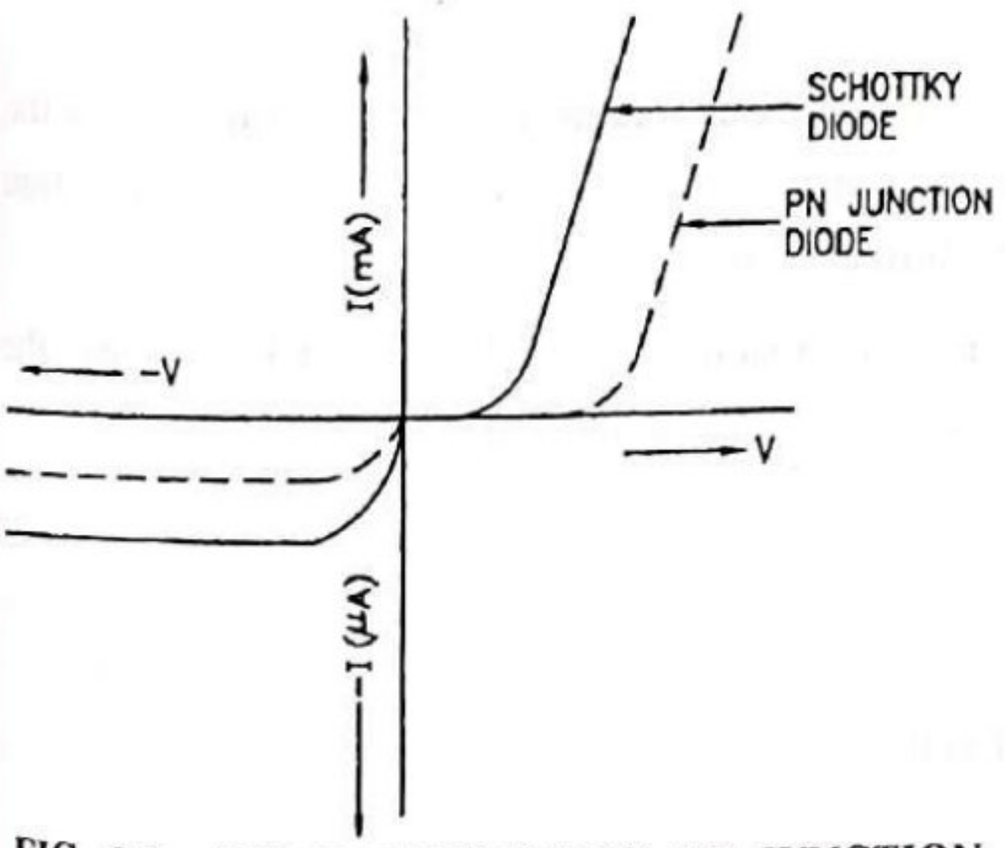


FIG. 3.9 : CHARACTERISTICS OF JUNCTION DIODE AND SCHOTTKY DIODE

Fig. 3.9 shows the forward and reverse characteristics of the conventional junction diode and the Schottky diode.

It is seen that, for the Schottky diode the cut in voltage is less compared to the PN junction diode. Also the magnitude of the reverse saturation current is higher than that of the junction diode.

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3.5.3 Applications :

Schottky diode has many useful applications. It is used as rectifier at high frequencies of the order of 300 MHz and more. Since it has zero noise figure, it is used in sensitive communication receivers and radars. It is also used in switched mode power supplies at the frequencies of 20 GHz. Due to its high switching speed it is useful in digital computers.

3.6 VARACTOR DIODE :

Varactor diode is a diode the junction capacitance of which can be varied by varying its reverse bias. Varactor diode is also known as the voltage variable capacitor diode or varicap or epicap or voltacap.

We know that a parallel plate capacitor is formed when a dielectric material is kept between two conducting plates. Capacitance so formed is directly proportional to the area A of the plates and inversely proportional to the distance d between the plates.

$$C \propto \frac{A}{d}$$

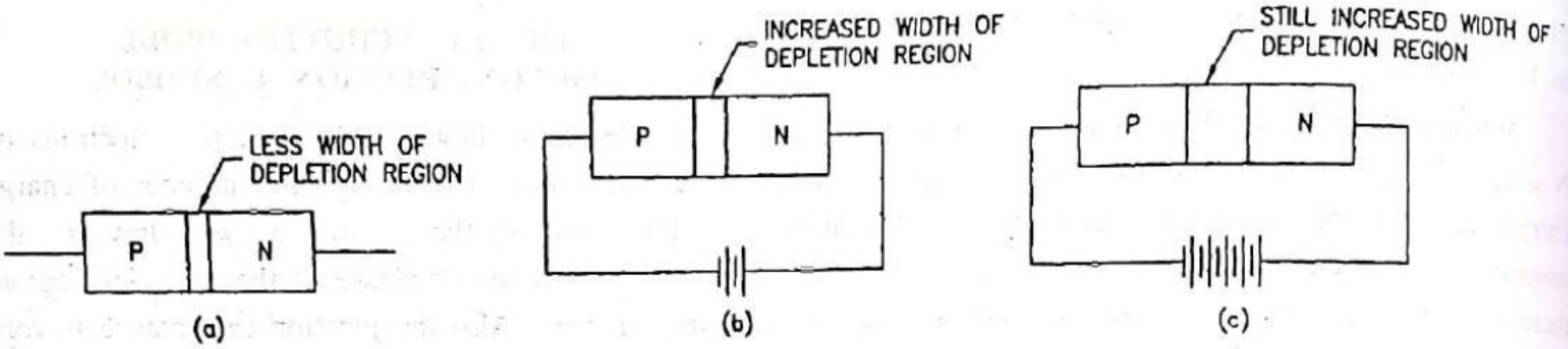


FIG. 3.10 : (a) PN JUNCTION WITHOUT BIAS (b) PN JUNCTION WITH SMALL REVERSE BIAS (c) PN JUNCTION WITH MORE REVERSE BIAS

When PN junction is formed, holes from P-region go to the N-region and electrons from N-region go to the P-region. Thus N-region near junction gets positive charge and P-region gets negative charge. Movement of charge carriers then stops and the depletion region is formed. The region in which there are no charge carriers is called the depletion region. This means that the depletion region behaves as an insulator. On one side the positive charge is stored and on another side the negative charge is stored. So the junction capacitance is formed. Width of the depletion region is small when no bias is given to the diode. Value of the junction capacitance is about 20 pF.

When PN junction is given some reverse bias, the width of the depletion region increases (Fig. b). So the value of the junction capacitance decreases. When the reverse bias is increased further the width of the depletion region increases further and the junction capacitance decreases further (Fig. c).

Thus varactor diode is the reverse biased diode whose junction capacitance can be varied by varying the reverse voltage. In this the doping density and doping profile are precisely controlled.

Capacitance is given by

$$C = \frac{\varepsilon A}{W}$$

Where ε is the permittivity of the semiconductor material,

A is the area of cross section of the junction

W is the width of the space charge region.

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Now W, the width of the space charge region is approximately proportional to the square root of the reverse bices voltage V since ε and A are constant for a particular diode,

$$C \propto \frac{1}{\sqrt{V}}$$

$$\therefore \quad C = \frac{K}{\sqrt{V}}$$

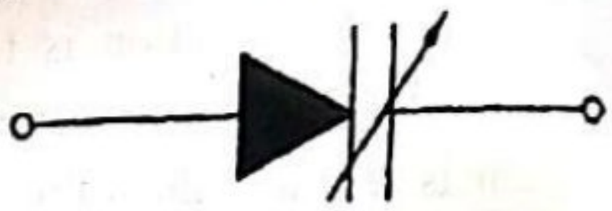


FIG. 3.11 : SYMBOL OF VARACTOR DIODE

Thus by varying the reverse voltage, the capacitance can be varied. Value of the capacitance can be varied between 6 pF to 550 pF. Symbol of the varactor diode is shown in Fig. 3.11.

3.6.1 Characteristic :

In Fig. 3.12 is shown the characteristic of varactor diode. This characteristic shows the relation between the reverse bias and the capacitance.

It is seen that the value of capacitance decreases with the increase in the reverse bias. For example when the reverse bias is 2 V, the value of capacitance is 380 pF, but it decreases to 110 pF when the reverse bias is increased to 8 V.

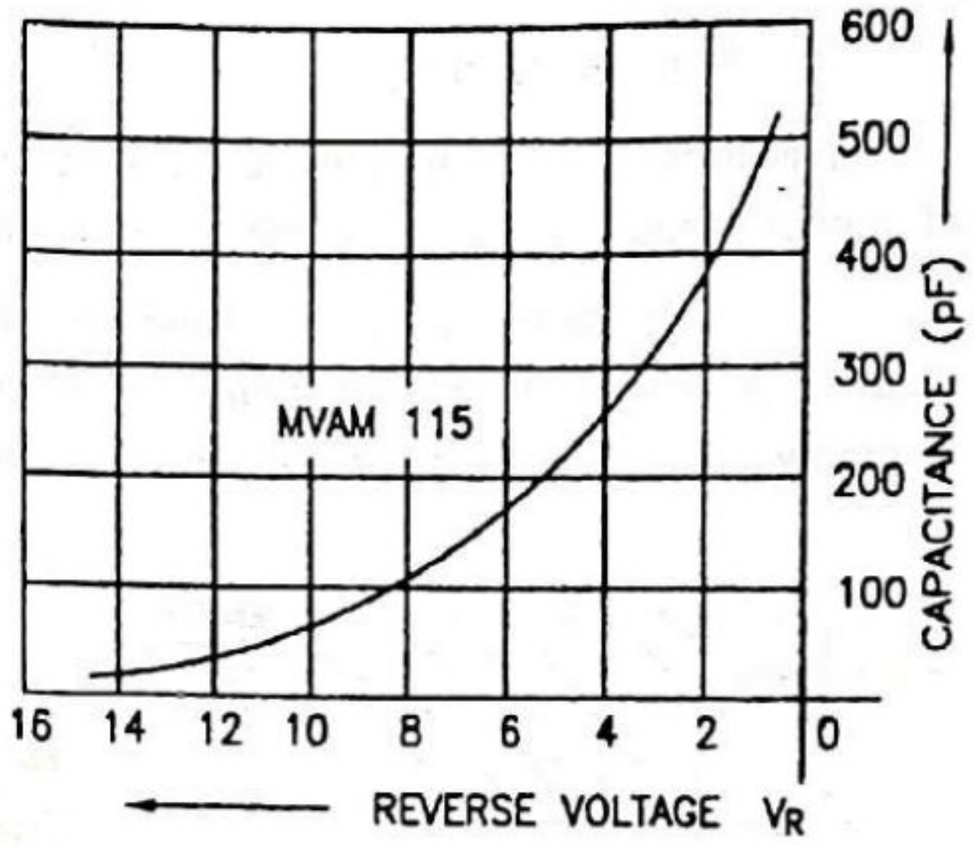


FIG. 3.12 : CHARACTERISTIC OF VARACTOR DIODE

3.6.2 Applications :

Various applications of varactor diode are as below :

1. It is used in voltage type tuning. In this by changing reverse voltage, capacitance is changed. So the resonant frequency is changed.
2. It is used in self balancing bridge circuit.
3. It is used in parametric amplifiers which are special types of amplifiers.
4. It is used in frequency modulation circuits.

3.6.3 Equivalent Circuit :

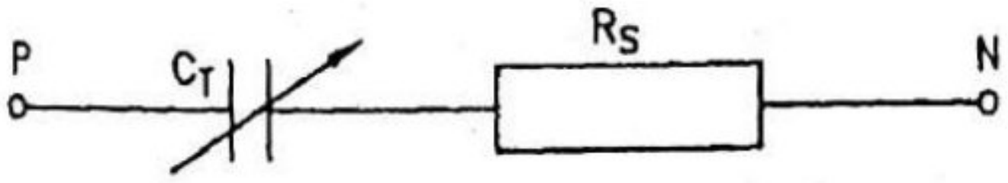


FIG. 3.13 : EQUIVALENT CIRCUIT OF A VARACTOR DIODE

Approximate equivalent circuit of varactor diode is shown in Fig. 3.13.

R_S is the resistance of the semiconductor material and C_T is the capacitance of the diode.

3.6.4 Some Type Numbers :

Some type numbers of varactor diode are as shown below :

TIV 308, MVAM 109, MVAM 115, MVAM 125 etc.

3.7 TUNNEL DIODE :

Tunnel diode is also known as the Esaki diode from the name of its inventor Leo Esaki. This type of diode was invented in 1958. It is one junction, two terminal, negative resistance type semiconductor device. It is used in amplifier, oscillator and switch at high frequencies.

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3.7.1 Principle of Operation :

When PN junction is formed, depletion region is formed near the junction. Width of the depletion region is dependent on the concentration of the impurity. That means it is dependent on doping. Width of the depletion region is less and the value of the potential barrier is less when the doping level is high. Normally the impurity is in proportion 10^8 in 1 where-as it is kept as 10^3 in 1 in tunnel diode. It means that in tunnel diode heavy doping is done. So as the width of the depletion region is less, barrier potential is less. So when small forward or reverse bias is given (less than the barrier potential), the charge carriers get sufficient energy and cross the junction and can go from one side to the other. This process of crossing the charge carries from one side to the other at low bias is called tunneling and this type of diode is called the tunnel diode.

3.7.2 Working of Tunnel Diode with the Help of Energy Band Diagram :

It is necessary to know the effect of doping on the energy band diagram before understanding the working of tunnel diode. In Fig. 3.14 energy band diagram of normally doped N or P-type semiconductor is shown.

In this, the holes in valence band are filled by electrons and the conduction band is empty. When P-type material is heavily doped, the number of electron is less so the valence band is not completely filled. So there is empty energy band above the valence band. Fig. 3.15(a).

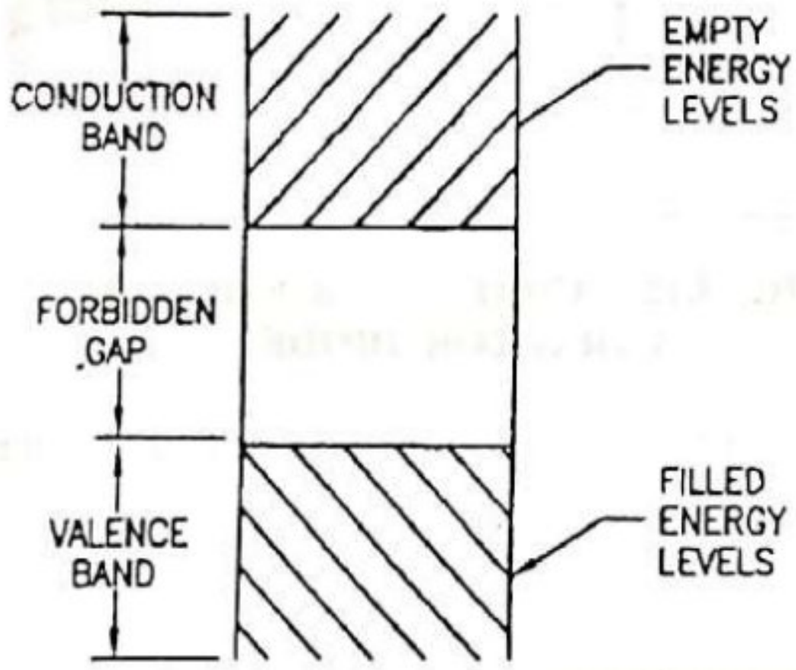


FIG. 3.14 : ENERGY BAND DIAGRAM OF NORMALLY DOPED N OR P-TYPE SEMICONDUCTOR

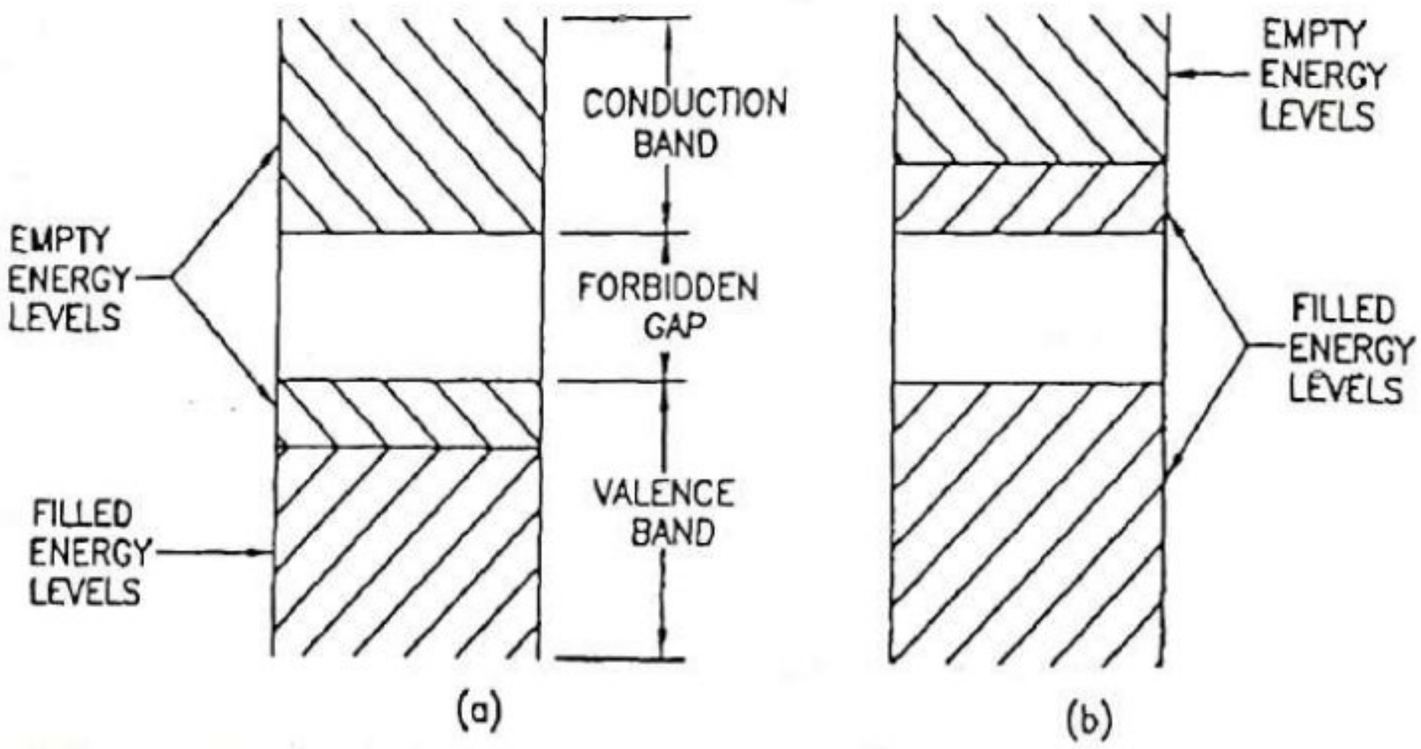


FIG. 3.15 : HEAVILY DOPED (a) P-TYPE MATERIAL (b) N-TYPE MATERIAL

Similarly when N-type material is heavily doped, there are more number of electrons. So electrons fill the valence band and produce filled energy level below the conduction Band (3.15(b)). Now let us try to understand the effect on energy band diagram when the tunnel diode is reverse biased and forward biased. In Fig. 3.16 is shown the energy band diagram of tunnel diode when no bias given.

It is seen that the width of the depletion region is very narrow and energy bands of P and N-type materials are similar to that shown in Fig. 3.15(a) and (b) and the filled levels on P and N-side are opposite to each other.

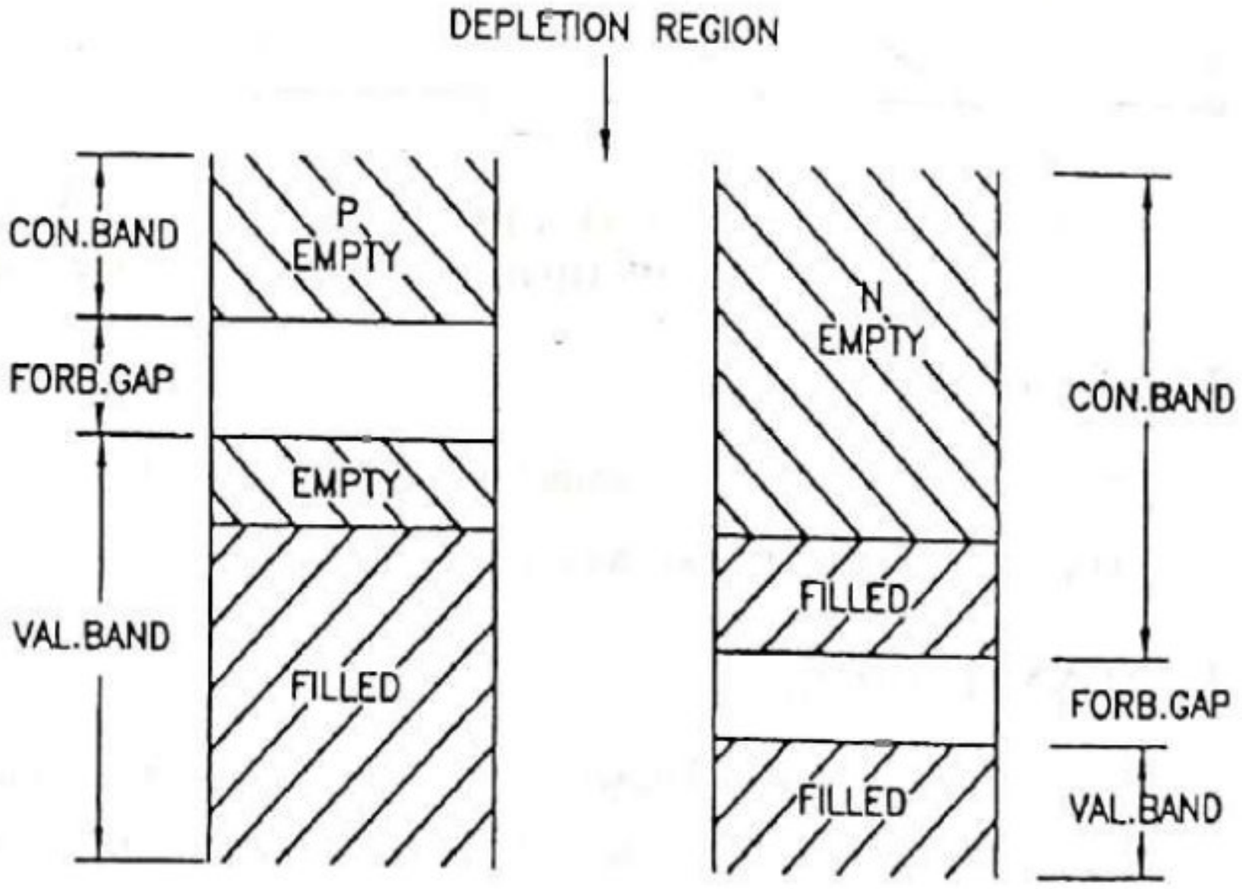


FIG. 3.16 : TUNNEL DIODE WITHOUT BIAS

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Now let us apply a small reverse bias to the junction. Electrons from P-side go from high energy level to the low energy level on N-side crossing the junction (tunneling occurs) and sufficient reverse current flows even the junction is reverse biased – OR in Fig. 3.17(c).

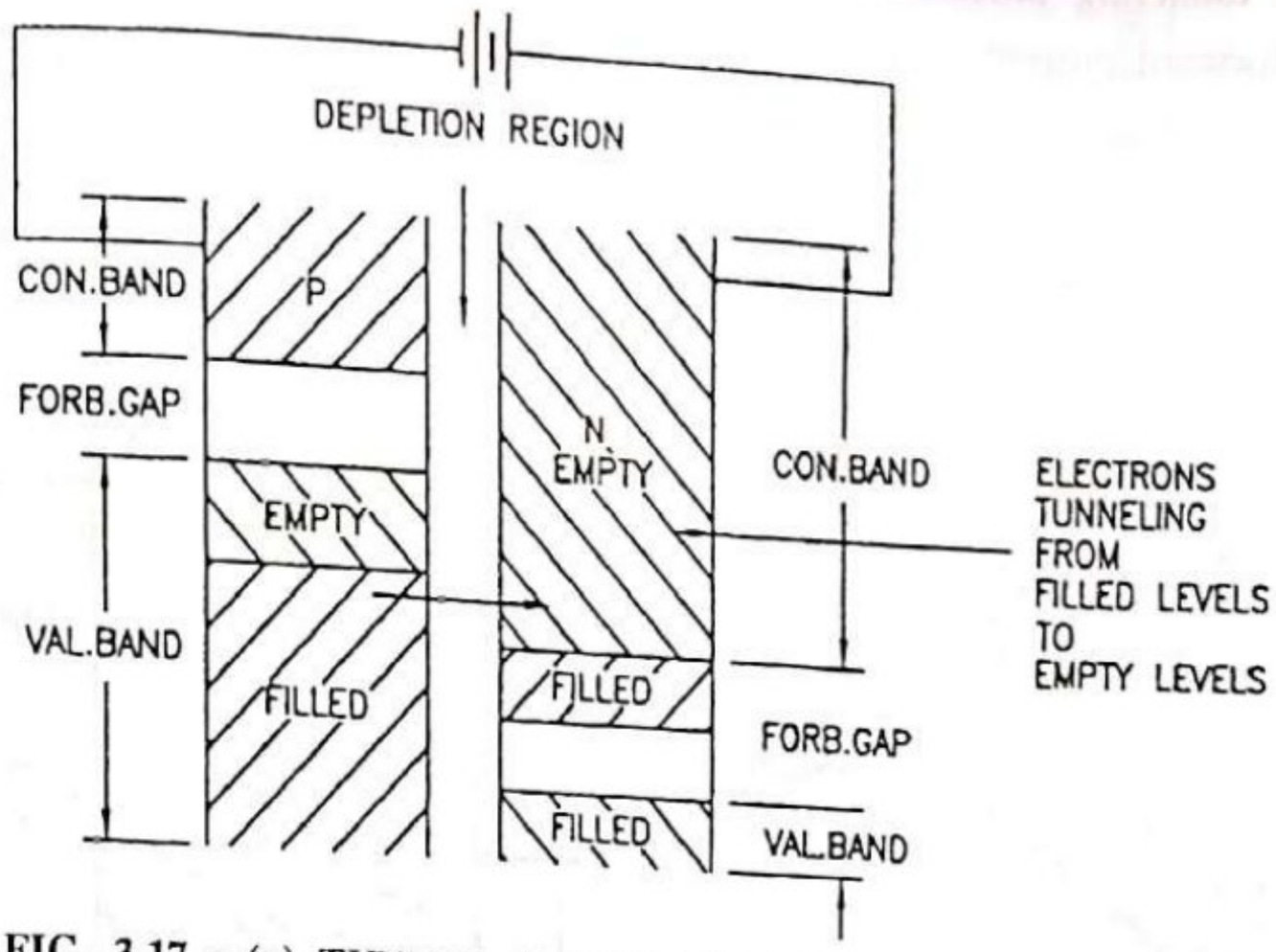
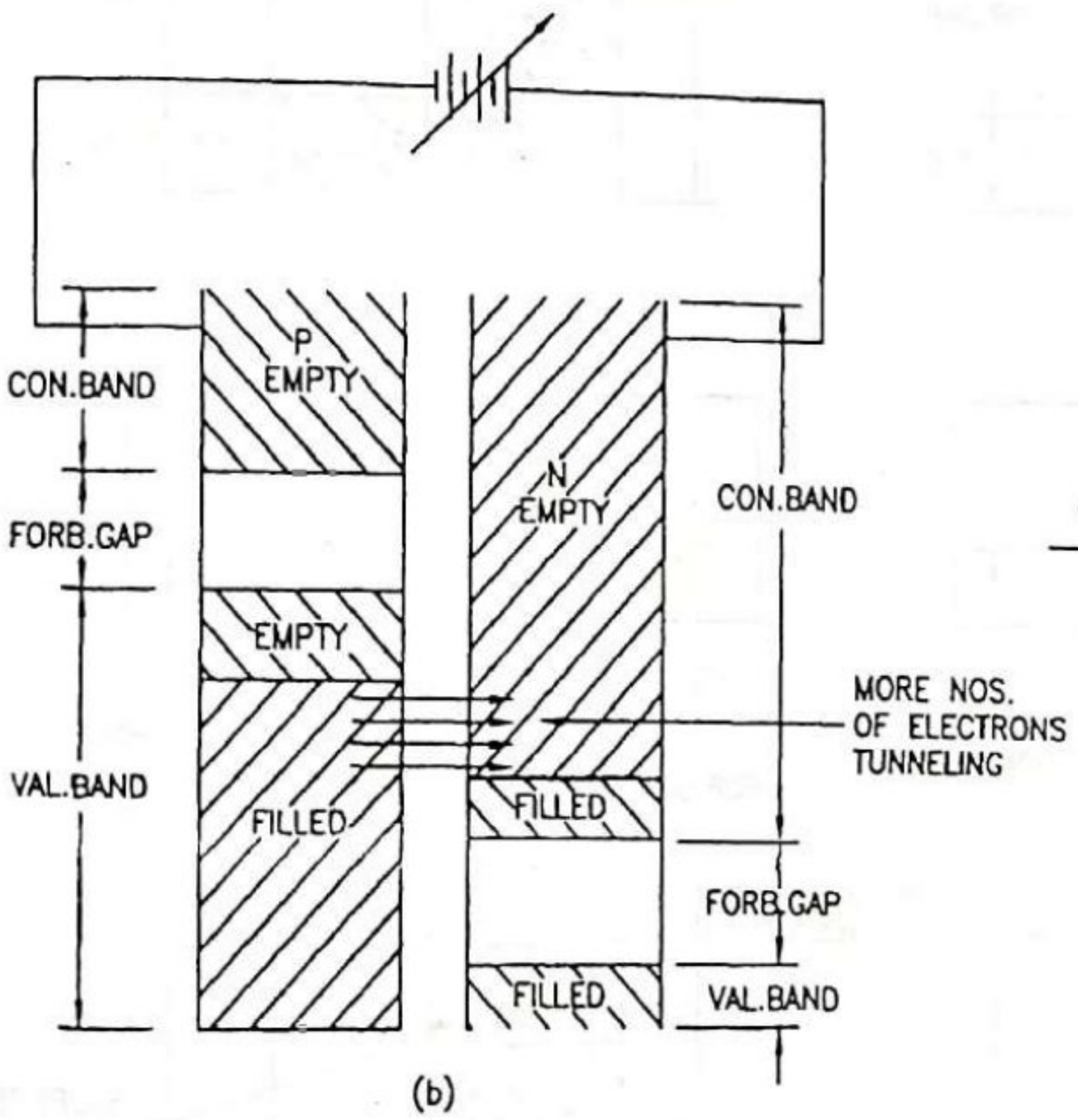


FIG. 3.17 : (a) TUNNEL DIODE WITH SMALL REVERSE BIAS



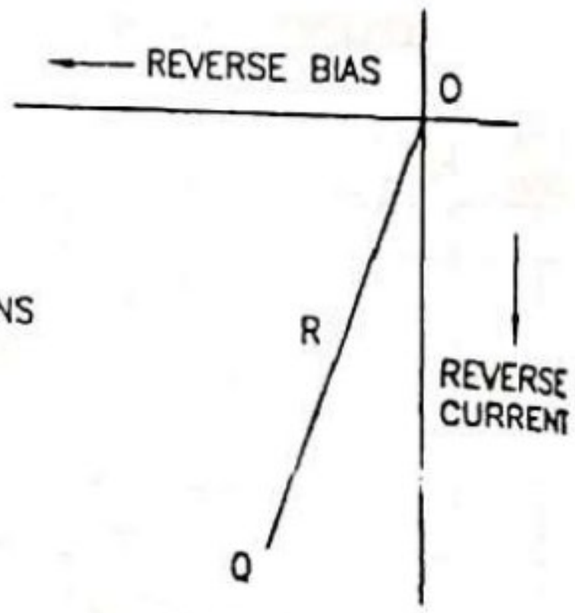


FIG. 3.17 : (b) WITH MORE REVERSE BIAS (c) REVERSE CHARACTERISTIC

When the reverse bias is increased, more and more electrons move from P to N side due to the tunneling action and reverse current increases-PQ. Thus reverse characteristic of tunnel diode is linear.

Now let us consider the behaviour when the tunnel diode is forward biased. When it is forward biased, filled level of N-type becomes more than the energy level of P-type. Due to this, tunneling of electrons occurs from N-type to P-type and we get forward current OA. Fig. 3.18 (a) and (e). Forward current increases with the increase in the forward bias. When the empty level above the valence band of P-type comes opposite to the filled level

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of N-type material, tunneling becomes maximum and maximum forward current flows (Fig. 3.18(b) and OB in Fig. 3.18(c)). Now when the forward bias is increased, some filled level of N-type comes opposite to the forbidden level of P-type and the tunneling process becomes slow (Fig. 3.18(c)). Less number of electrons travel from N to P-region and the forward current decreases (BC). And when the filled level of N material comes exactly opposite to the forbidden level of P-type material, the tunneling process stops (Fig. 3.18(d)) and the forward current becomes minimum (D).

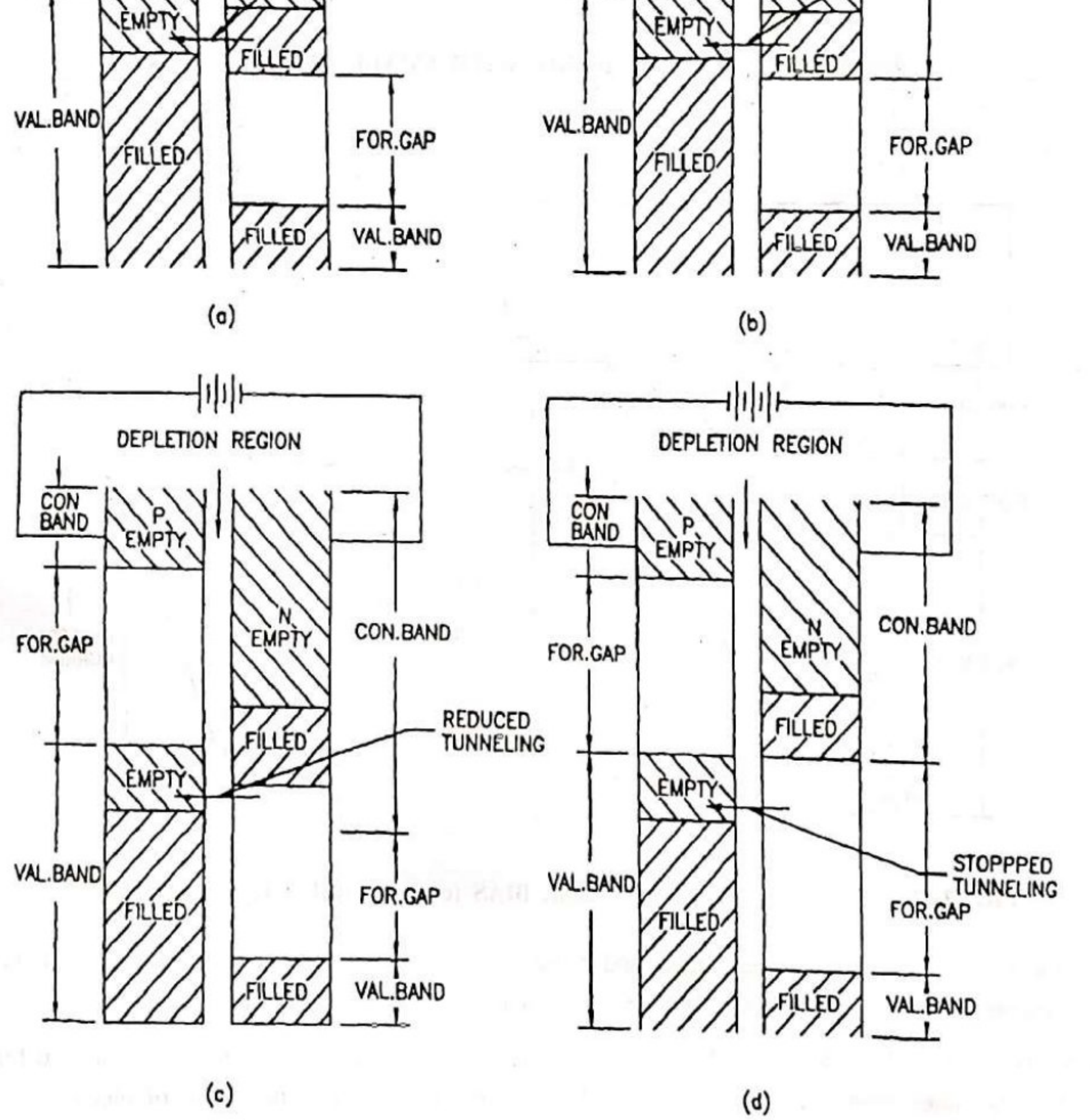


FIG. 3.18 : (a) TUNNEL DIODE FORWARD BIASED (b) MAXIMUM TUNNELING (c) REDUCED TUNNELING (d) MINIMUM FORWARD CURRENT

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Now if the forward bias is increased further, the tunnel diode behaves as ordinary junction diode, i.e. when the value of the forward bias exceeds the barrier potential, the forward current increases rapidly (DEF).

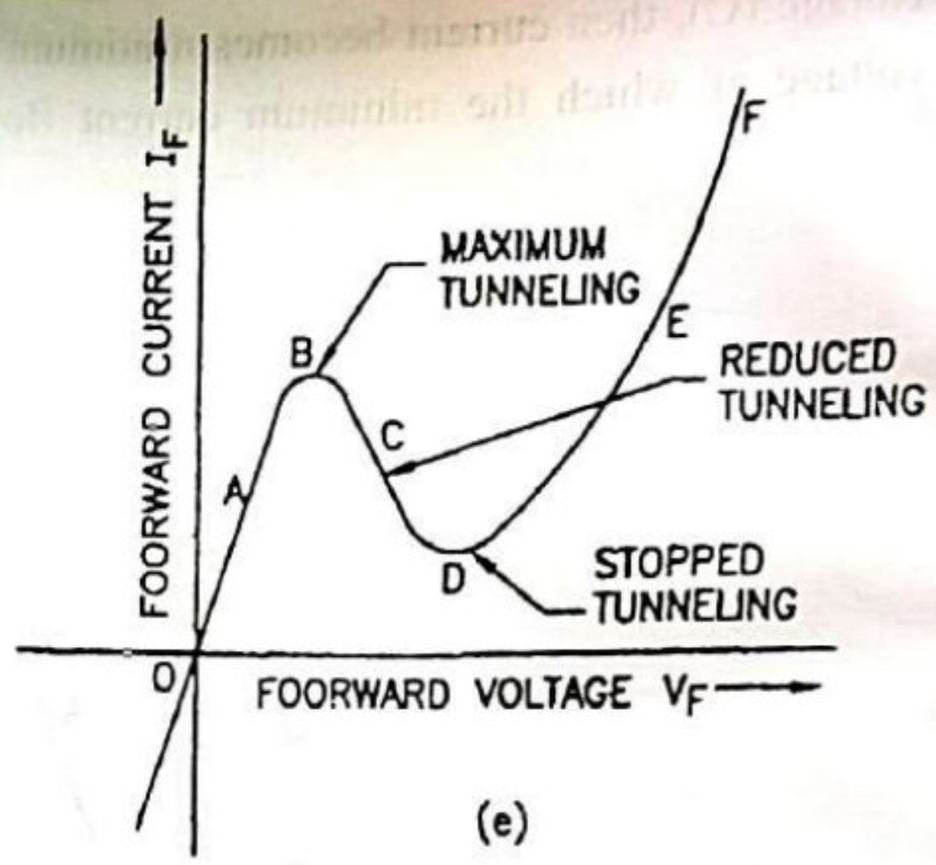


FIG. 3.18 : (e) CHARACTERISTIC

3.7.3 Method of Obtaining Characteristic :

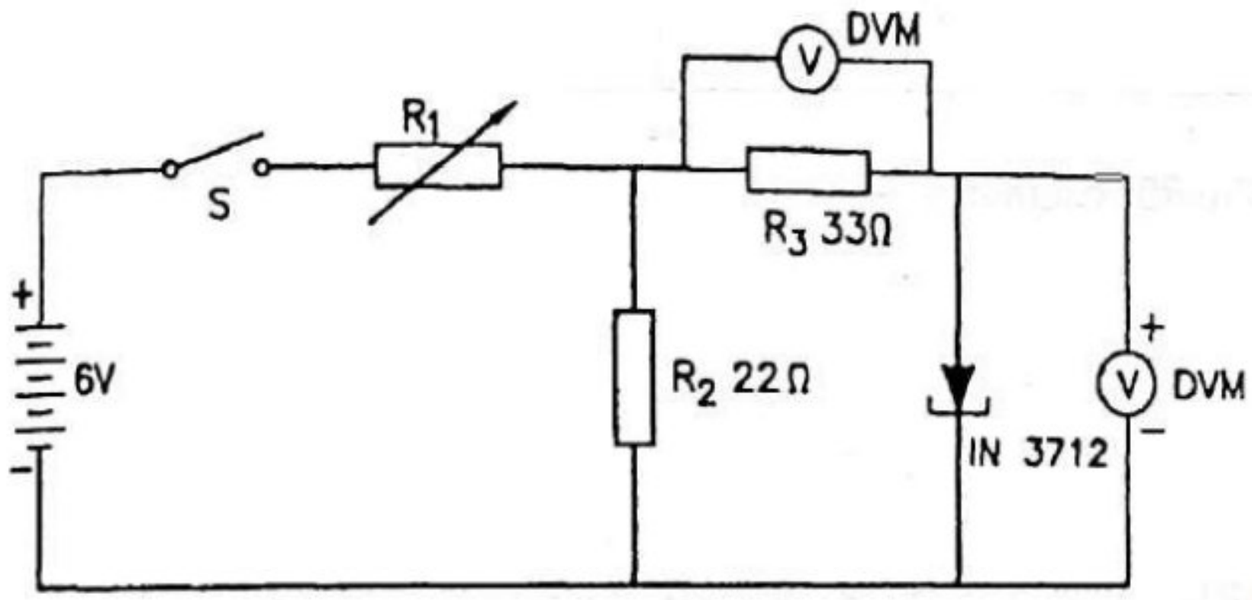


FIG. 3.19 : CONNECTION DIAGRAM TO OBTAIN THE CHARACTERISTIC OF TUNNEL DIODE

In Fig. 3.19 is shown the circuit arrangement of obtaining characteristic of the tunnel diode. In tunnel diode the important characteristic is before the barrier potential. The supply should be given from the millivolt source. If the millivolt source is not available, then the arrangement shown in Fig. 3.19 should be used. In this a variable resistance R_1 (decade resistance box) and a low resistance R_2 (say 22 Ω) are connected across the 6 V or 9 V source. So R_2 becomes low impedance source.

To measure the current flow, the voltage drop across the 33 ohm resistor (R_3) is measured using DVM.

$$\text{Current through diode} = \frac{\text{Voltage drop across } R_3}{\text{Value of } R_3}$$

Voltage across the diode is measured by DVM.

Connections are made as shown. Forward voltage is changed by changing the value of the decade resistance Rl in step. Forward voltage and voltage drop across R_3 are measured.

Forward current is calculated.

Reverse characteristic is not important but if it is desired the readings are taken as above after reversing the connections of the diode.

3.7.4 Characteristics :

In Fig. 3.20 are shown the reverse and forward characteristics of the tunnel diode.

From the reverse characteristic it is seen that considerable reverse current flows with less reverse voltage (OR). Sufficient forward current (OA) is also available with little forward voltage when the diode is forward biased. It increases with the increase in the forward voltage and becomes maximum (B). This maximum current is known as the peak current (I_P). The forward voltage at which the peak current is obtained is known as the peak voltage and is denoted by V_P. After the peak current I_P is reached, the current decreases with the increase in the forward

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voltage (C), then current becomes minimum (D). This minimum current is called the valley current I_V. The forward voltage at which the minimum current flows is known as the valley voltage V_V.

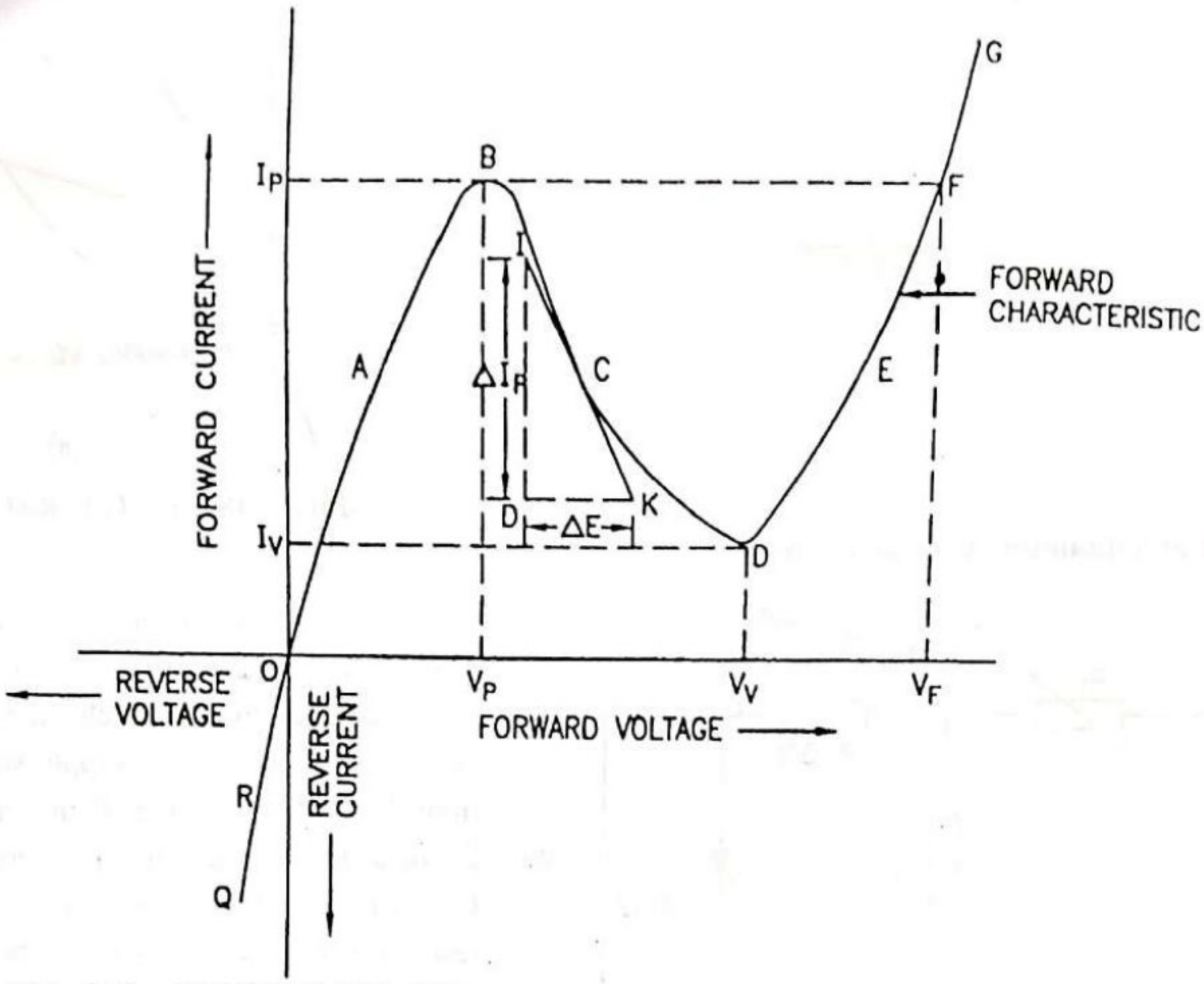


FIG. 3.20 : REVERSE AND FORWARD CHARACTERISTICS OF TUNNEL DIODE

Afterwards the forward current increases with the increase in forward voltage (DEFG) as in the conventional junction diode. At F, the value of forward current becomes equal to I_P. This is called the forward current and the forward voltage at which the value of the forward current becomes equal to the peak current I_P is known as the forward voltage V_F. Tunnel diode becomes totally forward biased at this voltage.

Normally when voltage is applied to a resistor the current through it increases with the increase in the voltage i.e. it obeys Ohm's law. But if we carefully look at the; characteristic, we find that in the portion BCD (i.e. between peak current I_P and valley current I_V) the current decreases with the increase in the voltage. This is called the negative resistance characteristic. This is the important characteristic of the tunnel diode. Because due to this negative resistance characteristic the tunnel diode can be used in oscillator. Value of the negative resistance changes at every point between BCD. So the value of the negative resistance is calculated at the middle of BCD i.e. at point C.

$$\text{Negative resistance} = \frac{\Delta E}{\Delta I} = \frac{JK}{JI}$$

3.7.5 Parameters :

Parameters of tunnel diode are as shown below :

Peak current $I_P = 1$ to 100 mA

Peak voltage $V_P = 50$ to 200 mV

Valey current $I_V = 0.1$ to 5 mA

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Valey voltage V_V = 350 to 500 mV

Forward voltage V_F = 0.5 to 1 V

Negative resistance = −0.1 to −200 Ω

3.7.6 Equivalent Circuit :

Small signal equivalent circuit of tunnel diode biased in the negative resistance region is shown in Fig. 3.21.

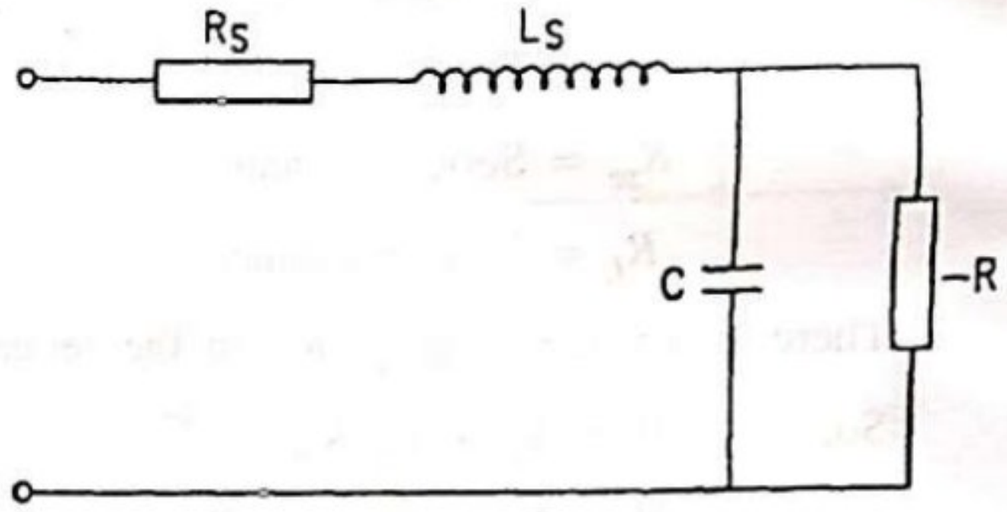


FIG. 3.21 : EQUIVALENT CIRCUIT OF A TUNNEL DIODE

R_S is the resistance of the semiconductor material and the load. Its value is between 1 to 5 Ω. Value of the inductance L_S is between 0.1 to 4 mH. Value of the shunt capacitance C is between 5 pF and 100 pF. Capacitor is in parallel with the negative resistance $-R$. Resonance frequency of tunnel diode due to L_S and C is between 700 MHz and 4 GHz.

3.7.7 Applications :

Various applications of tunnel diode are as below :

1. It is used as parallel amplifier.
2. It is used as series amplifier.
3. It is used as oscillator at high frequencies.
4. It is used as switch.

3.7.8 Difference Between Conventional Diode and Tunnel Diode :

Difference between the conventional diode and the tunnel diode are as under :

1. In conventional diode, the reverse current flows only after the breakdown voltage is reached i.e. reverse current is not available at the working voltage. Where-as in tunnel diode reverse current flows even with less reverse voltage.
2. In conventional diode sufficient forward current flows only when the value of the forward voltage exceeds the barrier potential. Where-as in tunnel diode, due to the tunneling effect sufficient forward current is available at the voltage well below the barrier potential.
3. In conventional diode, forward current increases with the increase in the forward voltage. Where-as in tunnel diode, in between the peak and valley current, forward current decreases with the increase in the forward voltage. This means it exhibits negative resistance characteristic.
4. Conventional diode can not be used as oscillator or amplifier. Where-as tunnel diode can be used as amplifier or oscillator.

3.7.9 Some Type Numbers :

Some type numbers of tunnel diode are as under :

IN 3712, IN 3714, IN 3128 etc.

3.8 ZENER DIODE VOLTAGE REGULATOR :

This type of voltage regulator is simple and less costly. It is used where good voltage regulation is not necessary. Basic circuit is shown in Fig. 3.22.

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Let, V_i = Input voltage

V_o = Output voltage

I_L = Load current

I_Z = Current through the zener diode

I_{se} = Total current = $I_L + I_Z$

R_{se} = Series resistance

R_L = Load resistance

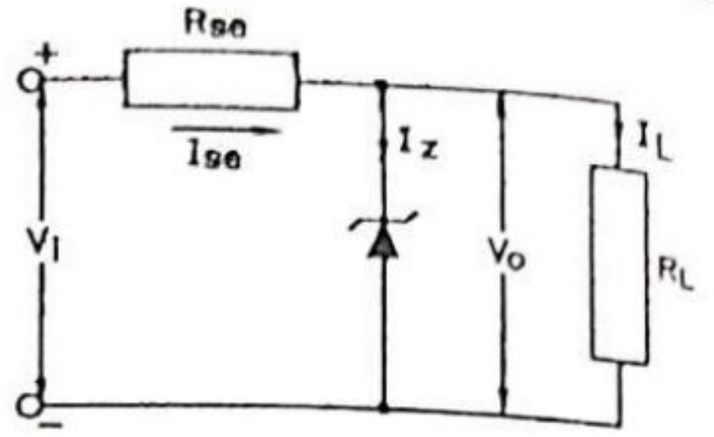


FIG. 3.22 : ZENER DIODE VOLTAGE REGULATOR

There is voltage drop $I_{se}\ R_{se}$ in the series resistance

So, $V_i = V_o + I_{se}\ R_{se}$

$V_i = V_o + (I_L + I_Z)R_{se}$

Vo is the zener voltage. Zener diode adjusts its current such that the output voltage (zener voltage) remains constant even if the output voltage tries to change due to the change in the input a.c. voltage or change in the load current.

If the load current I_L remains constant and if there is increase in the input voltage, the zener diode draws more current so I_{se} and hence $I_{se}\ R_{se}$ drop increases and the output voltage remains constant. Similarly if the load current remains constant and if the input voltage decreases, the zener diode draws less current so the $I_{se}\ R_{se}$ drop decreases to make output voltage constant.

Now when the input voltage remains constant and if the load current increases, the output voltage V_o does not decrease as zener diode takes less current so value of I_{se} and drop $I_{se}\ R_{se}$ remain constant. Similarly with the input voltage remaining constant and if the load current decreases the output voltage remains constant as zener diode takes more current to make I_{se} constant.

And when there is change in both the input voltage and the load current, the output voltage remains constant as zener adjusts its current accordingly.

3.8.1 Limitations of Zener Voltage Regulator :

(1) Temperature stability decreases at higher current i.e. there is decrease in the output voltage at higher temperature.

(2) Efficiency is reduced due to the power loss in zener diode.

(3) Voltage regulation is not very good.

(4) Zener current increases to high value when the load current reduces to zero so its power dissipation increases.

3.8.2 Application of Varactor Diode :

We have discussed that we can increase or decrease the junction capacitance of varactor diode simply by changing the reverse bias on the diode. This makes a varactor diode ideal for use in circuits that require voltage-controlled tuning. Fig. 3.23 shows the use of varactor diode in a tuned circuit. Note that the capacitance of the varactor is in parallel with the inductor. The varactor and the inductor form a parallel LC circuit. For normal operation, a varactor diode is always operated under reverse bias. In fact, this condition is met in the circuit shown in Fig. 3.23. The resistance R_W in the circuit is the winding resistance of the inductor. This winding resistance

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is in series with the potentiometer R_1. Thus R_1 and R_W form a voltage divider that is used to determine the amount of reverse bias across the varactor diode D_1 and therefore its capacitance. By adjusting the setting of R_1, we can vary the diode capacitance. This, in turn, varies the resonant frequency of the LC circuit. The resonant frequency f_r of the LC circuit is given by;

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

If the amount of varactor reverse bias is decreased, the value of C of the varactor increases. The increase in C will cause the resonant frequency of the circuit to decrease. Thus, a decrease in reverse bias causes a decrease in resonant frequency and vice-versa.

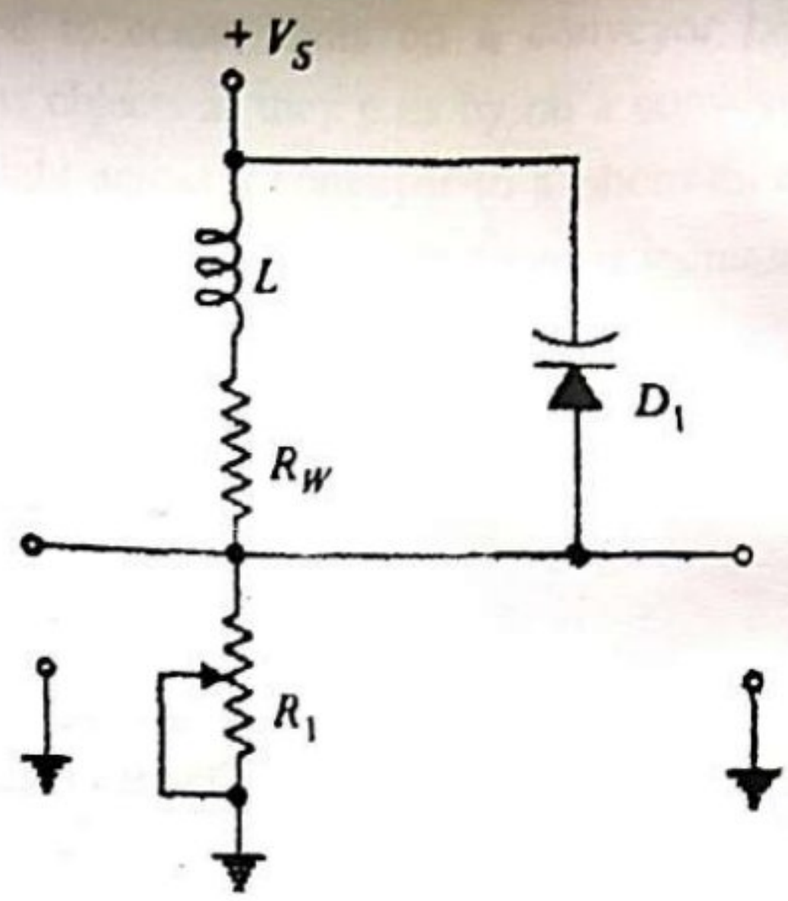


FIG. 3.23

3.8.3 Schottky Diode :

The schottky diode is a type of metal – semiconductor junction diode, which is also known as hot-carrier diode, low voltage diode or schottky barrier diode. The schottky diode is formed by the junction of a semiconductor with a metal. Schottky diode offers fast switching action and has a low forward voltage drop. As we are aware that in a PN junction diode, P-type and N-type are joined together to form a PN junction. Whereas, in a Schottky diode metals like platinum or aluminium are used instead of P type semiconductors.

1. Schottky Diode Symbol :

The symbol for the Schottky barrier diode is based around the basic diode circuit symbol. The circuit symbol of the Schottky diode is shown in the Fig. 3.24.

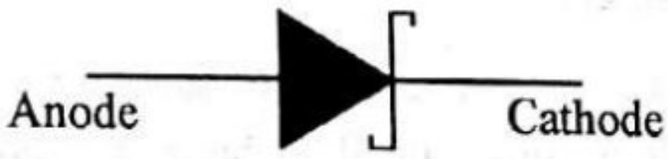


FIG. 3.24 : SCHOTTKY DIODE SYMBOL

2. Working of a Schottky Diode :

The operation relies on the principle that the electrons in different materials have different potential energy.

N-type semiconductors have higher potential energy than electrons of metals.

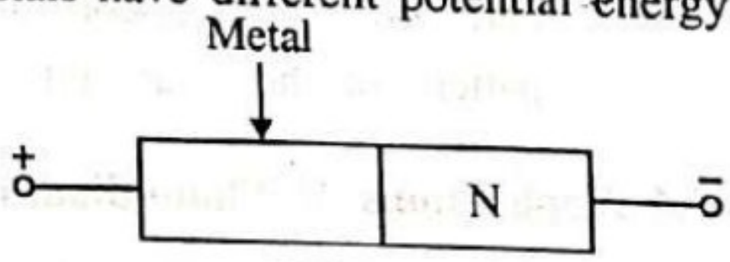


FIG. 3.25

When these two are brought into contact, there is a flow of electrons in both directions across the metal-semiconductor interface.

A voltage is applied to the Schottky so that the metal is positive when compared to the semiconductor.

The voltage opposes the built-in potential and makes the current flow easy.

3. V-I Characteristics of Schottky Diode :

The V-I characteristics of Schottky diodes are very much similar to the PN junction diode. Current is the dependent variable while voltage is the independent variable in the Schottky diode. The forward voltage drop of the Schottky diode is low between 0.2 to 0.3 volts.

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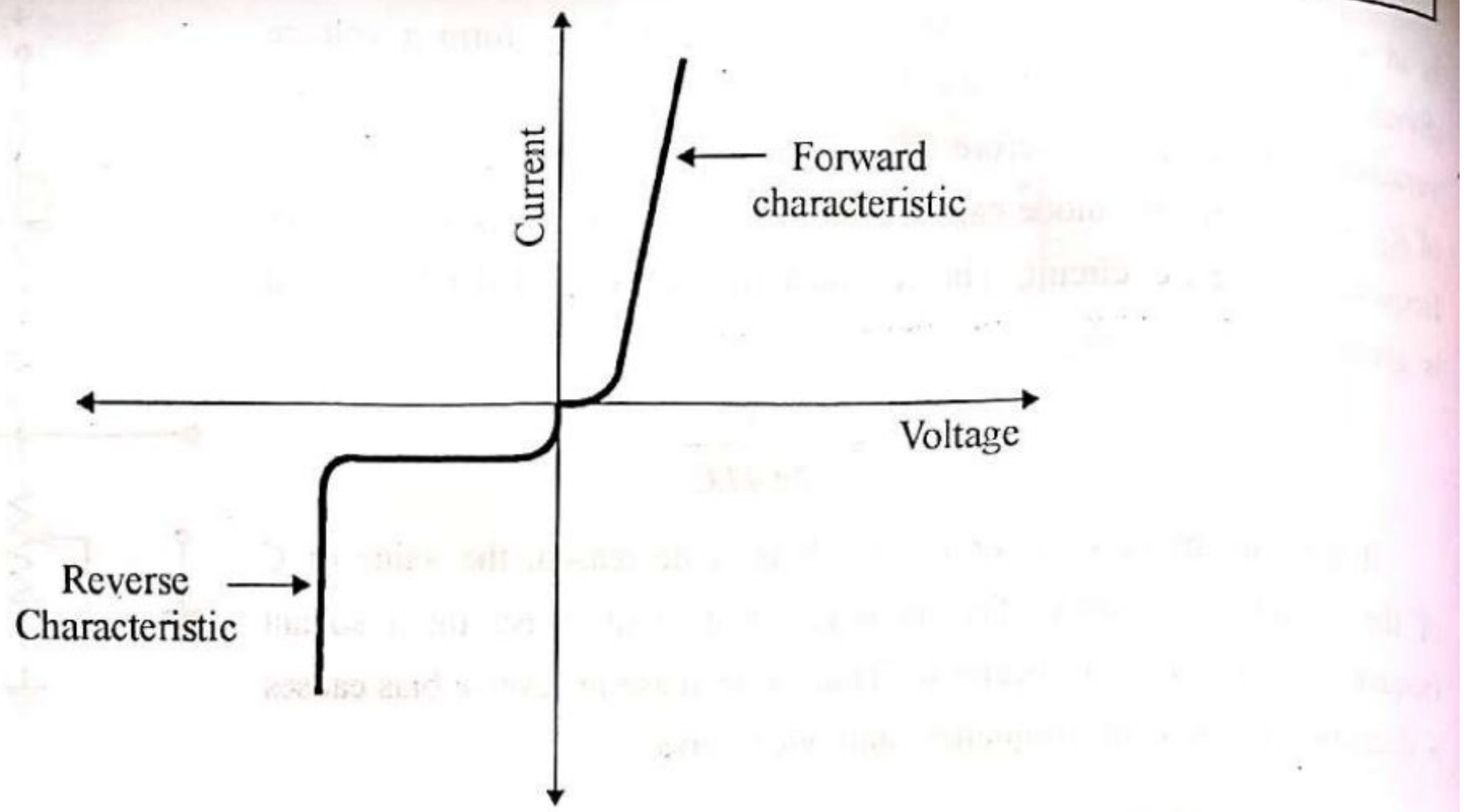


FIG. 3.26

4. Applications of Schottky Diode :

Schottky diodes have been useful for the industry of electronics that has spotted many applications in diode rectifiers because of its unique properties. Here are some major areas where it is widely used.

(1) **RF mixer and detector diode :** The Schottky diode consists of its radio frequency functions owing to its switching speed at the highest level and top frequency capability. The Schottky barrier diodes come handy for diode ring mixers with high performance.

(2) **Power rectifier :** The Schottky barrier diodes also have functions with high power as rectifiers. The high density of current and voltage drop with low forward shows that the wastage of power is lesser than the normal PN junction diodes.

(3) **Power OR circuits :** This diode would be useful for functions where two different power supplies drive a load like in battery supply. It is important that the power coming from supply should not mix with the others.

(4) **Solar Cell Applications :** As we know, the solar cells are usually linked to the batteries that are rechargeable, mostly batteries with lead-acid since power supply must be necessary round the clock. Solar cells would not support the applied charge in reverse and thus, a diode would be used in a proportional pattern of the solar cells.

3.8.4 Applications of Photo-diodes :

There are a large number of applications of photodiodes. However, we shall give two applications of photodiodes by way of illustration.

(1) Alarm circuit using photo-diode. Fig. 3.27 shows the use of photo-diode in an alarm system. Light from a light source is allowed to fall on a photo-diode fitted in the doorway. The reverse current I_R will continue to flow so long as the light beam is not broken. If a person passes through the door, light beam is broken and the reverse current drops to the dark current level. As a result, an alarm is sounded.

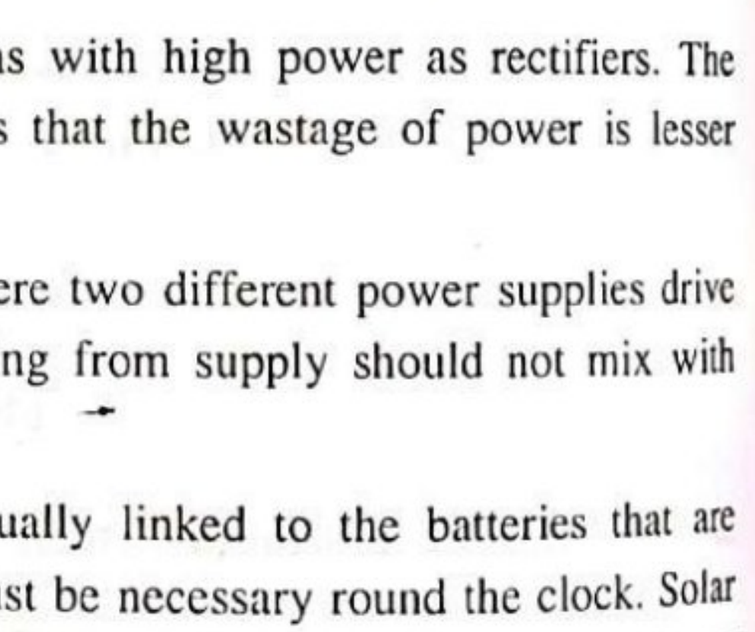


FIG. 3.27

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(2) Counter circuit using photo-diode. A photodiode may be used to count items on a conveyor belt. Fig. 3.28 shows a photo-diode circuit used in a system that counts objects as they pass by on a conveyor. In this circuit, a source of light sends a concentrated beam of light across a conveyor to a photo-diode. As the object passes, the light beam is broken, I_R drops to the dark current level and the count is increased by one.

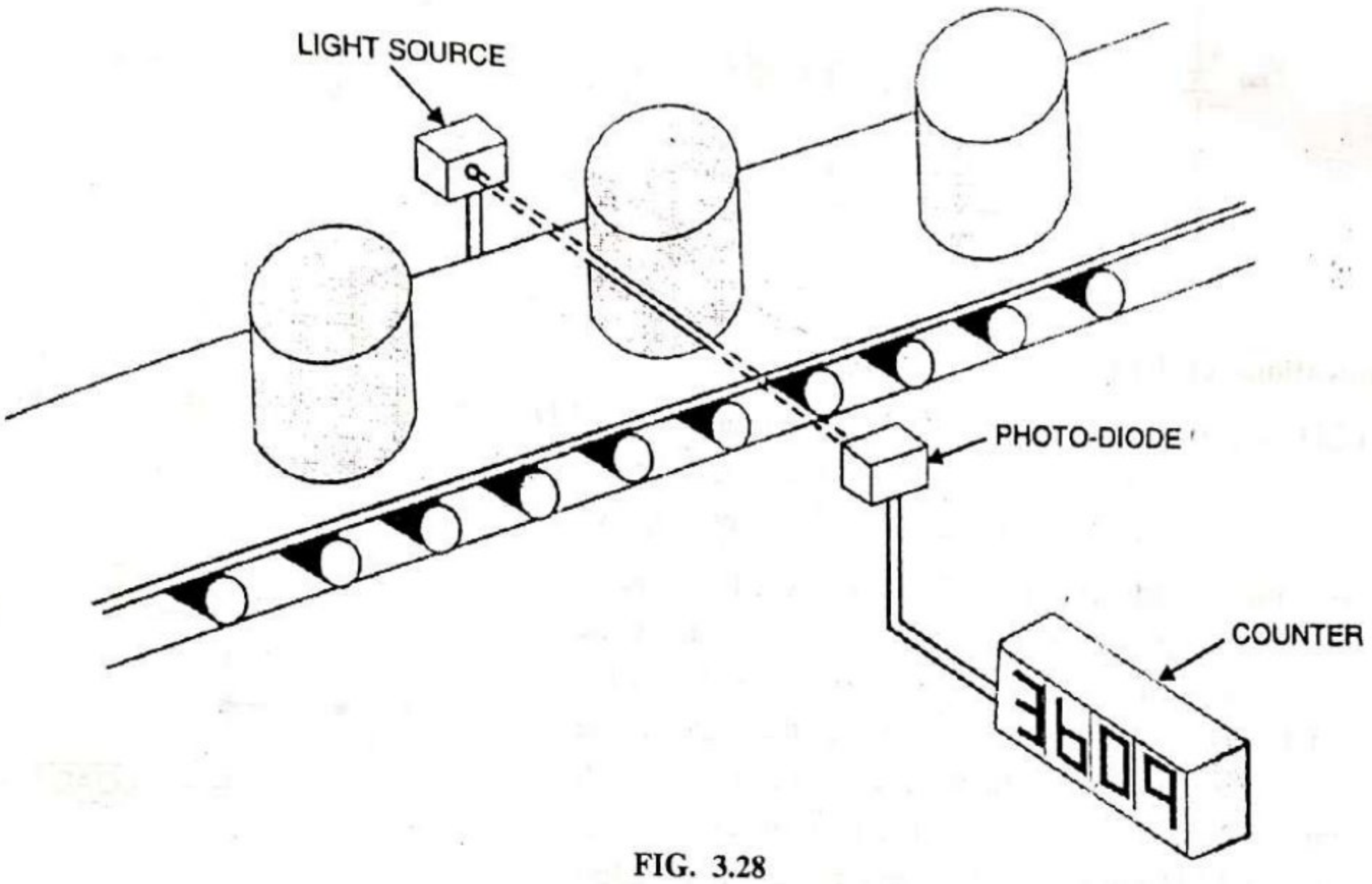


FIG. 3.28

3.8.5 Tunnel Diode Oscillator :

A tunnel diode is always operated in the negative resistance region. When operated in this region, it works very well in an oscillator. Fig. 3.29(i) shows a parallel resonant circuit. Note that R_P is the parallel equivalent of the series winding resistance of the coil. When the tank circuit is set into oscillations by applying voltage as shown in Fig. 3.29. (ii), damped oscillations are produced. It is because energy is lost in the resistance RP of the tank circuit.

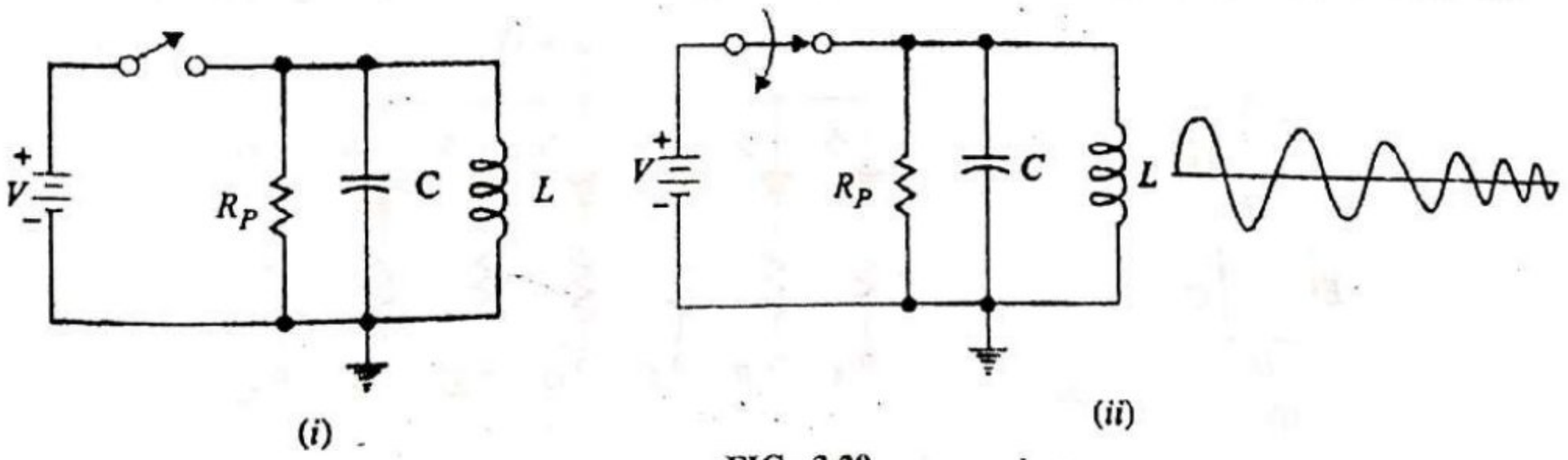


FIG. 3.29

If a tunnel diode is placed in series with the tank circuit and biased at the centre of the negative resistance portion of its characteristic as shown in Fig. 3.30, undamped oscillations are produced at the output. It is because the negative-resistance characteristic of the tunnel diode counteracts the positive-resistance characteristic of the tank circuit.

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The circuit shown in Fig. 3.30 is called tunnel diode oscillator or negative resistance oscillator. The negative resistance oscillator has one major drawback. While the circuit works very well at extreme high frequencies (upper mega hertz range), it cannot be used efficiently at low frequencies. Low-frequency oscillators generally use transistors.

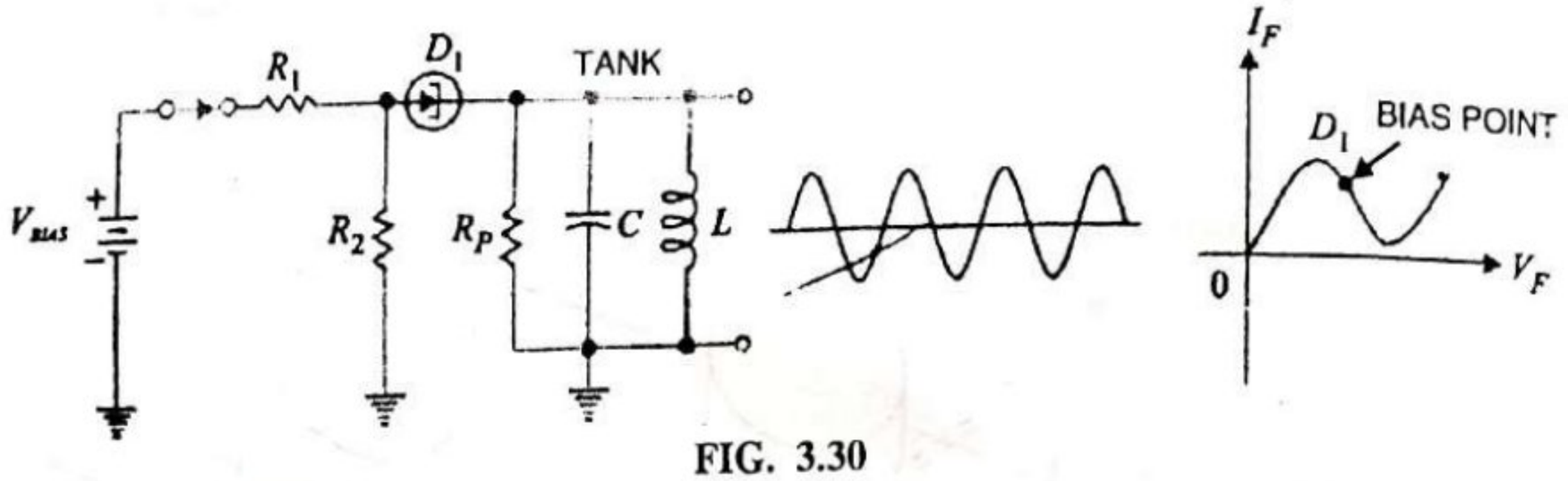


FIG. 3.30

3.8.6 Applications of LEDs :

The LED is a low-power device. The power rating of a LED is of the order of milliwatts. This means that it is useful as an indicator but not good for illumination. Probably the two most common applications for visible LEDs are (i) as a power indicator (ii) seven-segment display.

(1) **As a power indicator :** A LED can be used to indicate whether the power is on or not. Fig. 3.31 shows the simple use of the LED as a power indicator. When the switch *S* is closed, power is applied to the load. At the same time current also flows through the LED which lights, indicating power is on. The resistor RS in series with the LED ensures that current rating of the LED is not exceeded.

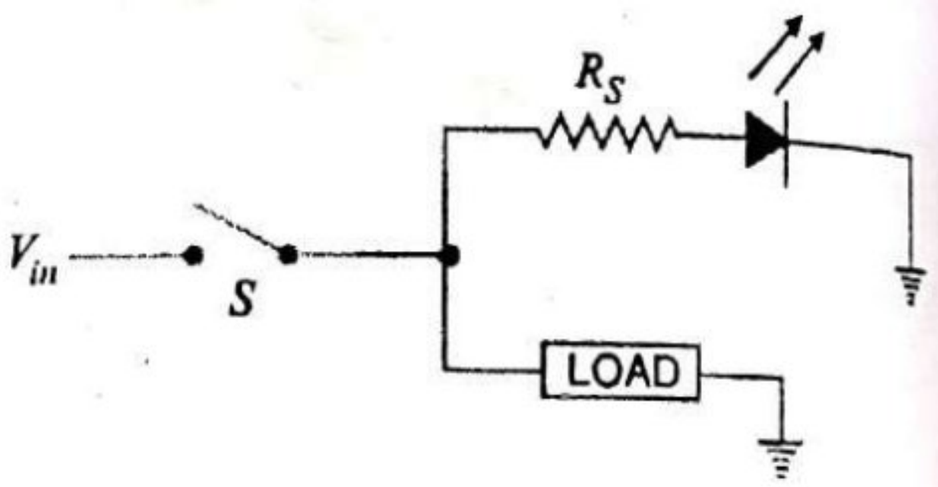


FIG. 3.31

(ii) **Seven-segment display :** LEDs are often grouped to form seven-segment display. Fig. 3.32 (i) shows the front of a seven segment display. It contains seven LEDs (A, B, C, D, E, F and G) shaped in a figure of 8. Each LED is called a segment. If a particular LED is forward biased, that LED or segment will light and produces a bar of light. By forward biasing various combinations of seven LEDs, it is possible to display any number from 0 to 9. For example, if LEDs A, B, C, D and G are lit (by forward biasing them), the display will show the number 3. Similarly, if LEDs C, D, E, F, A and G are lit, the display will show the number 6. To get the number 0, all segments except G are lit.

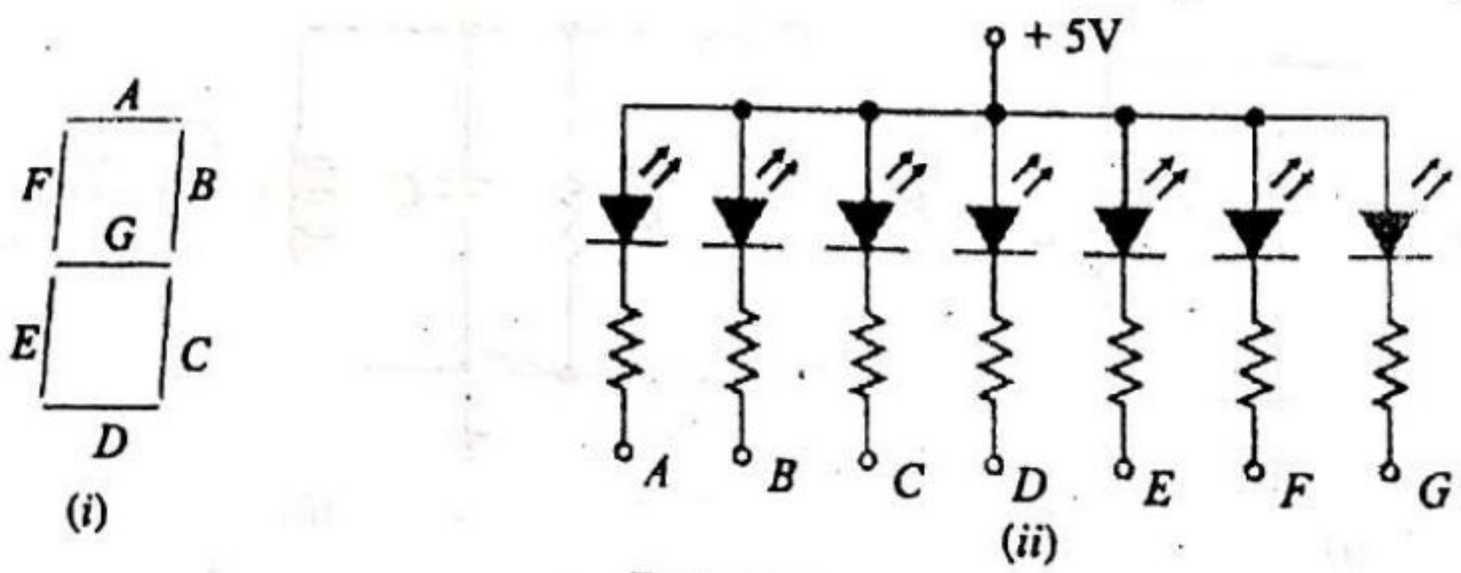


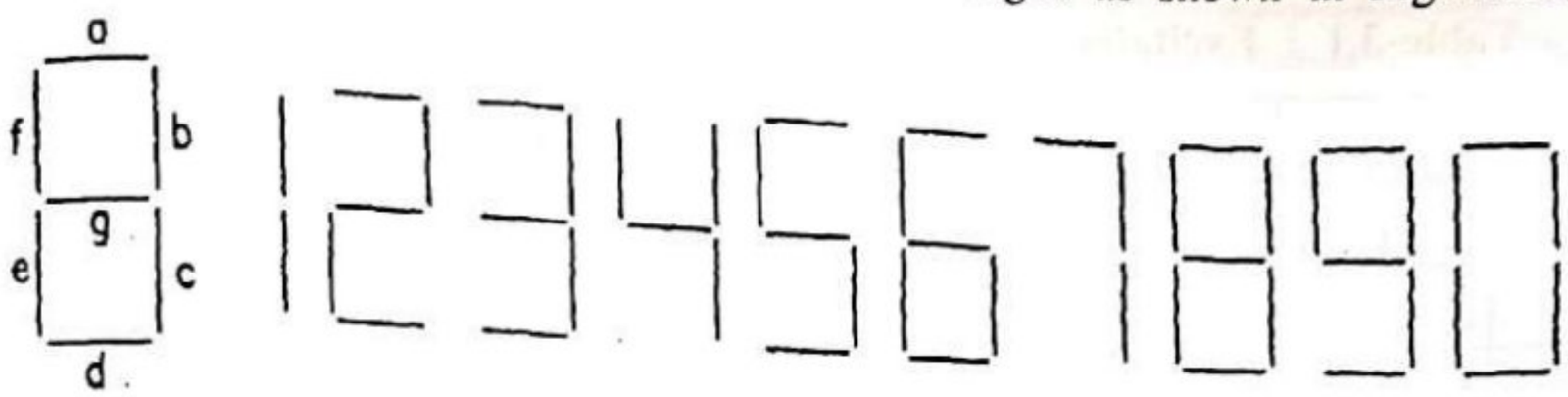
FIG. 3.32

Fig. 3.32(ii) shows the schematic diagram of seven-segment display. External series resistors are included to limit currents to safe levels. Note that the anodes of all seven LEDs are connected to a common positive voltage source of + 5 V. This arrangement is known as common-anode type. In order to light a particular LED, say A, we ground the point A in Fig. 3.32(ii). It forward biases the LED A which will be lit.

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3.8.7 Seven Segment LED Display :

In this seven rectangular type LEDs are fabricated and arranged as shown in Fig. 3.33(a).



(a) (b)

FIG. 3.33 : SEVEN SEGMENT LED DISPLAY (a) SEGMENT ARRANGEMENT (b) DIFFERENT NUMERALS

Seven segments are a, b, c, d, e, f and g. By exciting different segments, different numerals are followed. For example, if a, b, c, d and g are excited, number 3 is produced.

There are two types of seven segment type LED displays :

(1) Common cathode type and

(2) Common anode type.

Arrangement of seven segments for common cathode type display is shown in Fig. 3.34(a) and (b). In this cathodes of all the segments are connected together inside and common cathode lead is taken out. Seven anode leads are brought out. In addition one more segment for decimal is provided. Some times two-one on either side - are provided. The leads are connected to the pins of the dual in line package (DIP) similar to *IC*. Common cathode is connected to ground. The segment to be lit is connected to the positive supply through the current limiting resistor. These current limiting resistors are connected externally i.e. kept on PCB.

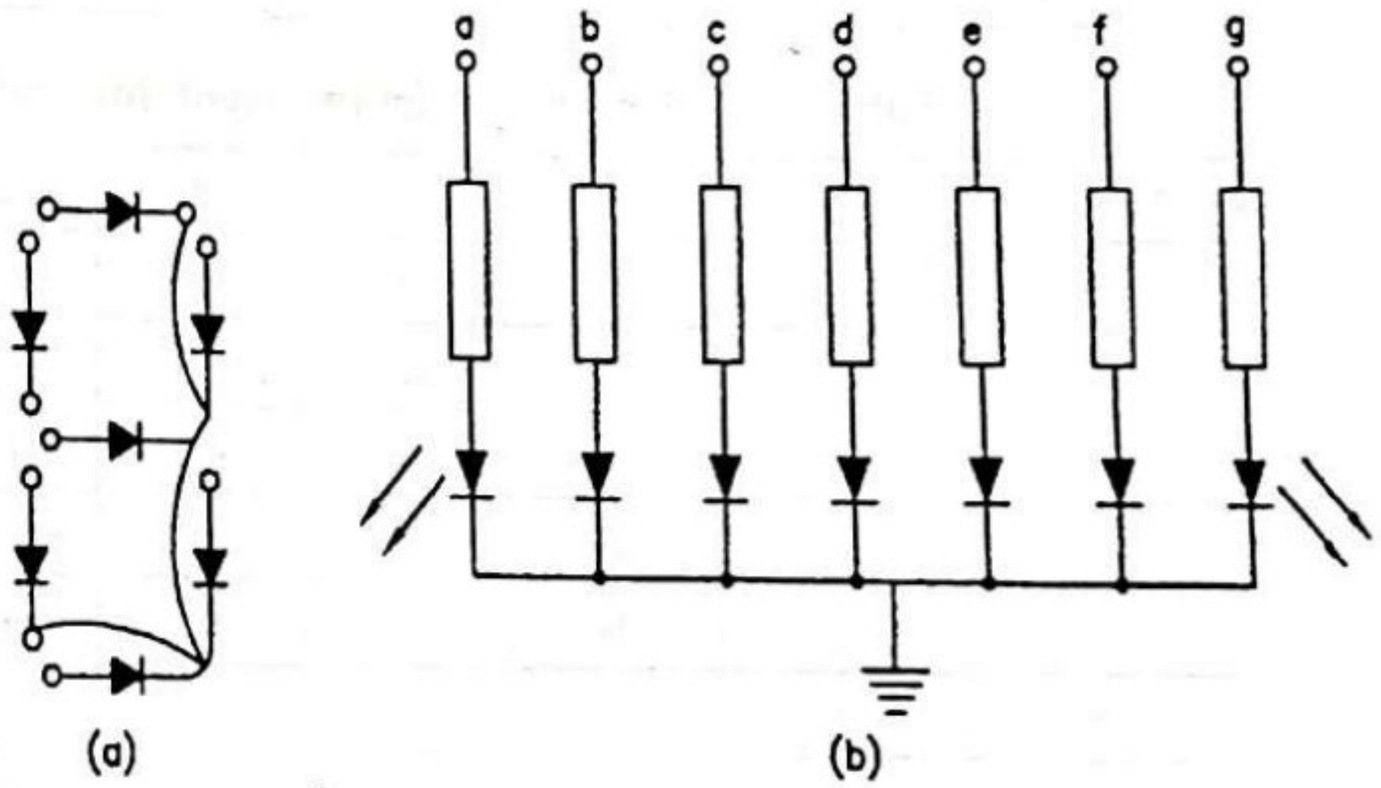


FIG. 3.34 : COMMON CATHODE SEVEN SEGMENT LED DISPLAY (a) SEGMENT ARRANGEMENT (b) CONNECTIONS

In common anode type display anodes of all the segments and decimal point are joined together and a common lead and cathode leads of the segments are brought out.

To excite the particular segment the common lead is connected to the positive supply and the segment to be lit is connected to the ground through the current limiting resistor.

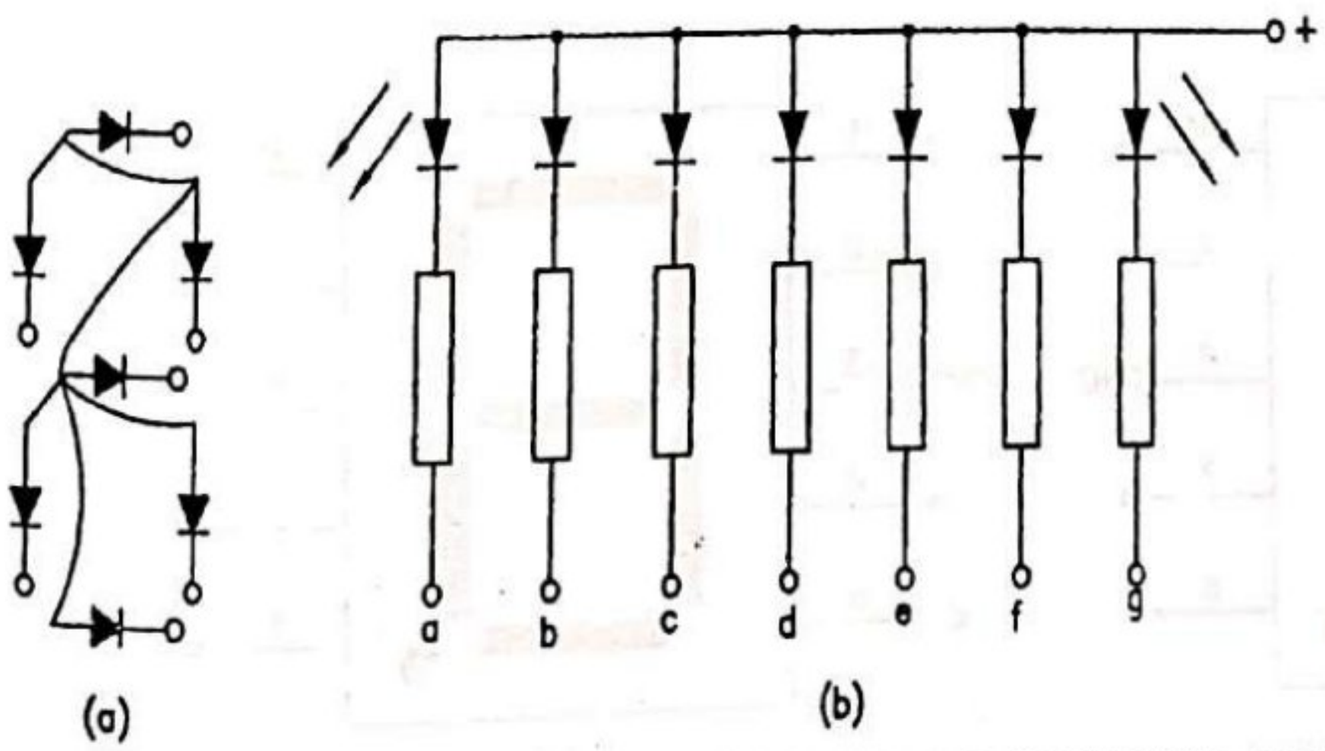


FIG. 3.35 : COMMON ANODE TYPE SEVEN SEGMENT LED DISPLAY (a) SEGMENT ARRANGEMENT (b) CONNECTIONS

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In Table No. 3.1 and 3.2, excitation arrangement is shown for common cathode and common anode type display respectively.

Table-3.1 : Excitation arrangement for common cathode type display

Number	a	b	c	d	e	f	g	Decimal Point
0	1	1	1	1	1	1	0	0
1	0	1	1	0	0	0	0	0
2	1	1	0	1	1	0	1	0
3	1	1	1	1	0	0	1	0
4	0	1	1	0	0	1	1	0
5	1	0	1	1	0	1	1	0
6	1	0	1	1	1	1	1	0
7	1	1	1	0	0	0	0	0
8	1	1	1	1	1	1	1	0
9	1	1	1	1	0	1	1	0

Table-3.2 : Excitation arrangement for common anode type display

Number	a	b	c	d	e	f	g	Decimal Point
0	0	0	0	0	0	0	1	1
1	1	0	0	1	1	1	1	1
2	0	0	1	0	0	1	0	1
3	0	0	0	0	1	1	0	1
4	1	0	0	1	1	0	0	1
5	0	1	0	0	1	0	0	1
6	0	1	0	0	0	0	0	1
7	0	0	0	1	1	1	1	1
8	0	0	0	0	0	0	0	1
9	0	0	0	0	1	0	0	1

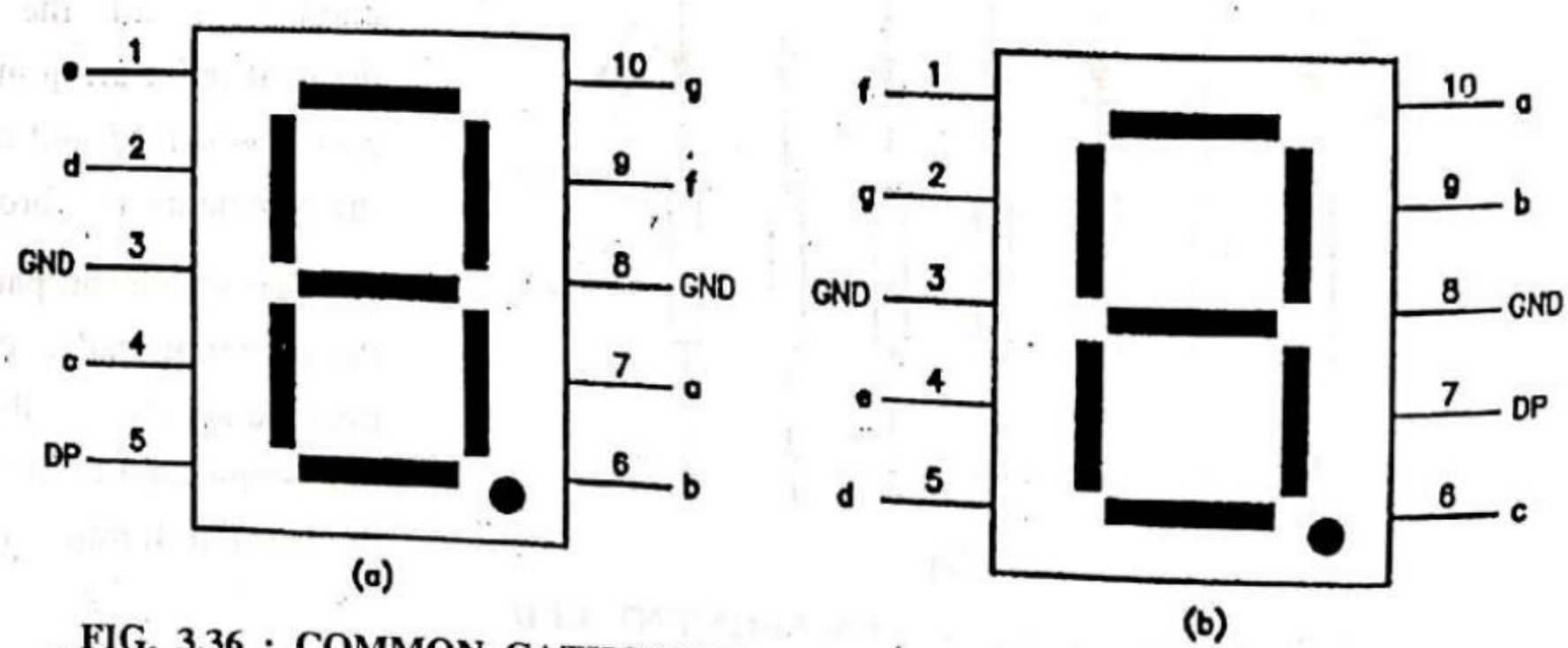


FIG. 3.36 : COMMON CATHODE DISPLAY (a) FND 503 (b) MAN 74 A

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3.9 CRYSTAL DIODE / POINT CONTACT DIODE :

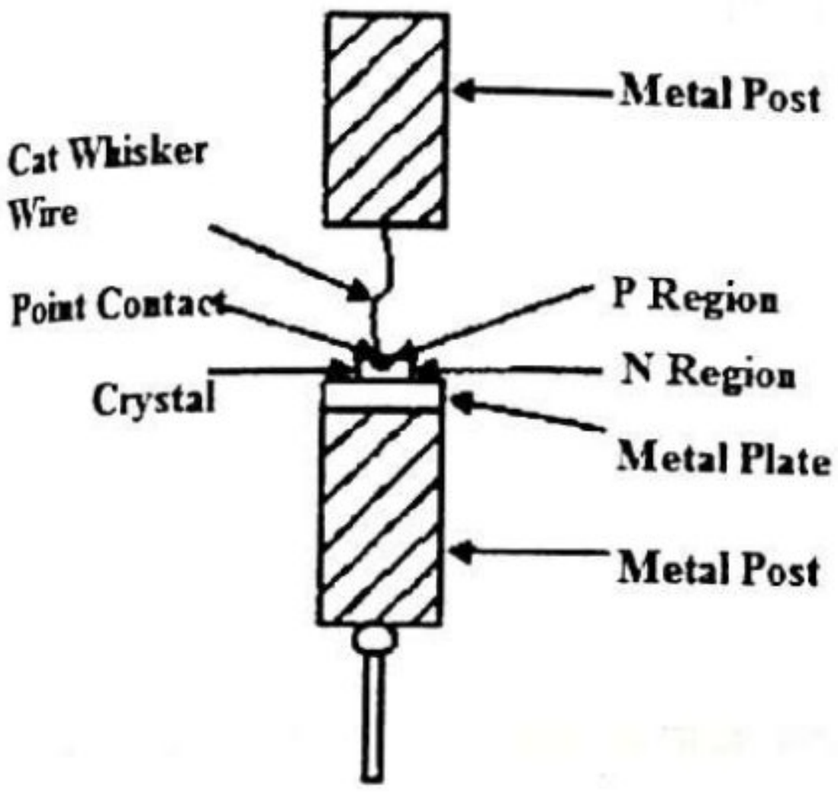
A crystal diode is also called a Cat's whisker because it is a type of point-contact diode. This diode operation mainly depends on the contact's force among the point and semiconductor crystal. This diode includes a metal wire that is pressed next to the semiconductor crystal. Here, the crystal acts works like a cathode and metal wire acts as an anode. These types of diodes are outdated within nature and used in detectors, microwave receivers, and rectifiers. The crystal diode symbol is shown below.



3.37 : CRYSTAL DIODE SYMBOL

Here, crystals are more popular semiconductors, so semiconductor materials are frequently known as crystals. This is how crystal triodes & diodes are named. These diodes are available in different types like common diodes which are mainly used for switching, rectifying, detecting, etc. and diodes including particular properties like voltage stabilizing, LEOs, photosensitive, etc.

3.9.1 Crystal Diode Construction :



3.38 : CRYSTAL DIODE / POINT CONTACT DIODE CONSTRUCTION

A crystal diode is designed with two dissimilar kinds of semiconductor materials beside the other. In this diode, one side is simple to flow the electrons whereas another side is not easy. But electrons try to jump from one side to the other across the border to form a current and nothing will take place. The alternating current which comes from outside can change by the diode to the direct current that most appliances require.

Generally, these diodes are also called point contact diodes which are the old type microwave semiconductor devices. These were mainly designed for utilizing in microwave receivers and at present, they are still used like detectors and receiver mixers. As compared to PN junction diode, crystal diodes mainly depend on the contact force among a semiconductor crystal and the point for its action.

Throughout this diode manufacturing, the huge current is supplied from the cat whisker to the Si crystal. The outcome of this huge current will form a small P-type material region in the region of the crystal within the surrounding area of the point contact so that a PN junction can be for Crystal Diode Characteristics.

The core of the crystal diode is the PN-junction and the main characteristic of this diode is unidirectional conductivity. The crystal diode's conductivity can be determined through the applied voltage across the diode & the flow of current throughout the diode. The main relationship between these two is known as the VI characteristics of the crystal diode. This diode characteristic can be divided into two main parts namely forward characteristics and reverse characteristics.

3.9.2 Forward Characteristic :

Once the forward voltage is applied across the diode then it increases gradually from zero. In the beginning, the forward current is approximately zero and this diode gives a high resistance which is known as the dead zone. The dead zone voltage of the Si diode is 0.5 V and the dead zone voltage of the Ge diode is 0.2 V.

When the diode's forward bias voltage is low as compared to the dead-band voltage; its forward current is zero. Once the applied voltage surpasses the dead-band voltage, the forward current starts to come into view until it is equivalent to the on-voltage and the forward current raises quickly. At this moment, the diode will be in the forward conduction position. The Sil tube's conduction voltage is 0.6 – 0.7 V and for the germanium tube is 0.2 to 0.3 V.

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3.9.3 Reverse Characteristic :

Whenever the reverse voltage is applied to the diode, the reverse current can be formed which is very small, and it does not change with the change of the reverse voltage within a large range. So this area is known as the reverse cut-off region.

The current that supplies once the reverse cut-off is known as reverse saturation current. Generally, silicon tubes include minimum microamperes to hundred microamperes whereas germanium tubes include a hundred microamperes to thousands of microamperes.

This current is a significant parameter to gauge the diode quality. When the value is small then the quality of the diode is better. Generally, the reverse saturation current can be ignored and in the reverse direction, the diode is considered non-conducting.

If the reverse voltage increases continuously to a fixed value, then the reverse current will increase suddenly. This incident is known as a reverse breakdown. At the moment, the applied voltage beyond the diode is known as the reverse breakdown voltage. This fact is not allowed once ordinary diodes are usually used. The diode is activated through a fixed forward voltage & reverse voltage cut-off characteristic which is known as unidirectional conductivity.

3.9.4 Advantages :

The advantages of crystal diode include the following :

1. These are more rugged, smaller.
2. Longer life.
3. Cheaper and Simpler.
4. They need no filament power. Consequently, they generate less heat as compared to the I equivalent vacuum diodes.

3.9.5 Disadvantages :

The disadvantages of crystal diode include the following :

1. These are very heat sensitive.
2. As compared to vacuum diodes, they can handle low inverse voltages and small currents.
3. They cannot place an overload even for a small phase.

3.9.6 Crystal Diode Application :

1. Crystal Diode Rectifier :

The electric power which is obtained from outside is mainly used for transmission that is AC; however, most of the devices work by using DC supply. As a result, we need to change AC to DC by using a rectifier circuit. Most of the rectifiers employ this diode through some resistors which are called crystal diode rectifiers. These are divided into two types like HWR Dr half-wave rectifier and FWR or full-wave rectifier.

The half-wave rectifier (HWR) conducts current simply in the positive half cycle of the applied input voltage In this rectifier, the AC supplies negative half cycles are suppressed. As a result, no current will be conducted and no voltage shows beyond the electric load So flow current always in one direction throughout the load.

In a full-wave rectifier (FWR), the flow of current throughout the load will be in the same direction for both the semi cycles of input AC voltage supply.

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2. Crystal Diode Detector :

In 1900s, it is primarily used in a crystal radio set as a signal detector. Crystal surface makes contact with the fine metallic probe. Thus, point contact diode got a descriptive name as a cat whisker detector. These are obsolete and consists of a thin, sharpened metal wire acting as an anode and semiconductor crystal acting as the cathode. This anode thin metal wire called as the cat's whisker wire is pressed against the cathode crystal. These crystal diode detectors were developed in the early 1900s and used in finding the hot spot on the semiconductor material crystal cathode which is manually adjusted for the best radio wave detection.

SUMMARY

1. PN junction diode with different type and doping and material defines special semiconductor diodes like zener, LED, photo diode, schottky diode, PIN, tunnel, photo voltage cell, photo conductive cell, photo transistor etc.
2. Zener diode is reverse biased operated diode and is used as a voltage regulator, clipper and clamper.
3. LED is forward biased operated diode with application as indicator, seven segment and fourteen segment displays, multicoloured LED and sensors.
4. Photodiode with addition of some assembly and components will be useful for sensor, detector, wave shape generator, oscillator and many others.
5. Tunnel diode is forward and reverse biased diode with specific requirement of supply and used as oscillator for special applications.
6. Varactor diode generally works in reverse biased mode and used as variable capacitor in applications like tunnel oscillator and wave shape generator.
7. Schottky diode is forward and reverse biased diode operated as high speed switch and mostly used in digital electronics logic ICs.
8. Crystal/point contact diode used in high speed rectifier circuit and detector circuit.

MULTIPLE CHOICE QUESTIONS (MCQs)

1. Zener diodes are used primarily as _____.

(a) amplifiers (b) voltage regulators (c) rectifiers (d) oscillators

2. A pn junction that radiates energy as light instead of as heat is called a _____.

(a) LED (b) photo-diode (c) photocell (d) Zener diode

3. The capacitance of a varactor diode increases when reverse voltage across it _____.

(a) decreases (b) increases (c) breaks down (d) stores charge

4. To display the digit 8 in a seven-segment indicator _____.

(a) C must be lighted (b) G must be off

(c) F must be on (d) All segments must be lighted

5. A photo-diode is normally _____.

(a) forward-biased (b) reverse-biased

(c) Neither forward nor reverse biased (d) Emitting light

6. When the reverse voltage increases, the junction capacitance _____.

(a) decreases (b) stays the same (c) increases (d) has more bandwidth

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7. The device associated with voltage-controlled capacitance is a _____.

(a) LED (b) photo-diode (c) varactor diode (d) Zener diode

8. The varactor is usually _____.

(a) forward-biased (b) reverse-biased

(c) unbiased (d) in the breakdown region

9. When the light increases, the reverse current in a photo-diode _____.

(a) increases (b) decreases (c) is unaffected (d) none of the above

10. To display the digit 0 in a seven segment display _____.

(a) A must be lighted (b) F must be off

(c) G must be on (d) all segments except G should be lighted

11. In an unregulated power supply, if load current increases, the output voltage _____.

(a) remains the same (b) decreases (c) increases (d) none of the above

12. In an unregulated power supply, if input a.c. voltage increases, the output voltage _____.

(a) increases (b) decreases (c) remains the same (d) none of the above

13. A power supply which has a voltage regulation of _____ is unregulated power supply.

(a) 0% (b) 0.5% (c) 10% (d) 0.8%

14. Commercial power supplies have voltage regulation _____.

(a) of 10% (b) of 15% (c) of 25% (d) within 1%

15. An ideal regulated power supply is one which has voltage regulation of _____.

(a) 0% (b) 5% (c) 10% (d) 1%

16. A zener diode utilises _____ characteristic for voltage regulation.

(a) forward (b) reverse

(c) both forward and reverse (d) none of the above

17. Zener diode can be used as _____.

(a) d.c. voltage regulator only (b) a.c. voltage regulator only

(c) both d.c. and a.c. voltage regulator (d) none of the above

18. A zener diode is used as a _____ voltage regulating device.

(a) shunt (b) series (c) series-shunt (d) none of the above

19. As the junction temperature increases, the voltage breakdown point for zener mechanism _____.

(a) is increased (b) is decreased (c) remains the same (d) none of the above

20. The rupture of co-valent bonds will occur when the electric field is _____.

(a) 100 V/cm (b) 0.6 V/cm (c) 1000 V/cm (d) more than 105 V/cm

21. In a 15 V zener diode, the breakdown mechanism will occur by _____.

(a) avalanche mechanism (b) zener mechanism

(c) both zener and avalanche mechanism (d) none of the above

22. A zener diode that has very narrow depletion layer will breakdown by _____ mechanism.

(a) avalanche (b) zener

(c) both avalanche and zener (d) none of the above

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23. As the junction temperature increases, the voltage breakdown point for avalanche mechanism _____.
(a) remains the same (b) decreases (c) increases (d) none of the above

24. Another name for zener diode is _____ diode.
(a) breakdown (b) voltage (c) power (d) current

25. Zener diodes are generally made of _____.
(a) germanium (b) silicon (c) carbon (d) none of the above

26. For increasing the voltage rating, zeners are connected in _____.
(a) parallel (b) series-parallel (c) series (d) none of the above

27. In a zener voltage regulator, the changes in load current produce changes in _____.
(a) zener current (b) zener voltage
(c) zener voltage as well as zener current (d) none of the above

28. A zener regulator _____ in the power supply.
(a) increases the ripple (b) decreases the ripple
(c) neither increases nor decreases ripple (d) data insufficient

29. When load current is zero, the zener current will be _____.
(a) zero (b) minimum (c) maximum (d) none of the above

30. The zener current will be minimum when _____.
(a) load current is maximum (b) load current is minimum
(c) load current is zero (d) none of the above

ANSWERS

1. (b) voltage regulators
2. (a) LED
3. (a) decreases
4. (d) All segments must be lighted
5. (b) reverse-biased
6. (a) decreases
7. (c) varactor diode
8. (b) forward-biased
9. (a) increases
10. (d) all segments except G should be lighted
11. (b) decreases
12. (a) increases
13. (c) 10%
14. (d) within 1%
15. (a) 0%
16. (b) reverse
17. (c) both d.c. and a.c. voltage regulator
18. (a) shunt
19. (b) is decreased
20. (d) more than 105 V/cm
21. (a) avalanche mechanism
22. (b) zener
23. (c) increases
24. (a) breakdown
25. (b) silicon
26. (c) series
27. (a) zener current
28. (b) decreases the ripple
29. (c) maximum
30. (a) load current is maximum

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EXERCISE

1. Draw and explain the characteristic of zener diode.
2. Write applications of zener diode.
3. State the specifications of zener diode.
4. Explain the construction of schottky diode and state is applications.
5. Explain the working of tunnel diode and draw the characteristics.
6. What is varactor diode ? Explain its working.
7. Draw and explain equivalent circuit of tunnel diode and method of obtaining the characteristics of tunnel diode.
8. State the advantages of photo diode.
9. Explain the construction of photo diode.
10. State advantages and applications of LED.
11. Explain the construction and working of varactor diode.
12. Explain the types of multicolored LED construction.
13. Compare photo diode with LED.
14. Explain the seven segment LED display.
15. Short note : Photo diode
16. Explain the use of varactor diode.
17. Explain constructions and working of LED.
18. Explain tunnel diode as oscillator.
19. Explain construction, working and characteristics of crystal diode.
20. Explain photodiode industrial application with suitable diagram.
21. State various applications of schottky diode.
22. Explain the construction and working of point contact/crystal diode.
23. State and explain application of crystal diode.

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